

Chapter 4: Motion in Two Dimensions



Motion in Two Dimensions



Helpful Videos

- Two-Dimensional Kinematics – Physics Concept Trailer:
<https://www.youtube.com/watch?v=shFNnyLztWE>
- How Do Horizontally Launched Projectiles Behave?
<https://www.youtube.com/watch?v=5-kZH97xr2U>
- Demonstrating the Components of Projectile Motion
<https://www.youtube.com/watch?v=2xCQ-MKRzlw>
- Professor Dave – Kinematics Part 3: Projectile Motion
<https://www.youtube.com/watch?v=aY8z2qO44WA>
- Professor Dave – Kinematics Part 4: Practice Problems and Strategy: <https://www.youtube.com/watch?v=1QbXXmyf2xc>

Chapter 4: Outline

4.1 Displacement and Velocity Vectors

4.2 Acceleration Vector

**Concepts &
Definitions**

4.3 2D Motion with Constant Acceleration

4.4 Projectile Motion

4.5 Uniform Circular Motion

**Physics Laws
& Formulas**

Chapter 4: Concepts and Definitions

Position vector

$$\vec{\mathbf{r}}(t) = x(t)\hat{\mathbf{i}} + y(t)\hat{\mathbf{j}} + z(t)\hat{\mathbf{k}}$$

Displacement vector

$$\Delta\vec{\mathbf{r}} = \vec{\mathbf{r}}(t_2) - \vec{\mathbf{r}}(t_1)$$

Velocity vector

$$\vec{\mathbf{v}}(t) = \lim_{\Delta t \rightarrow 0} \frac{\vec{\mathbf{r}}(t+\Delta t) - \vec{\mathbf{r}}(t)}{\Delta t} = \frac{d\vec{\mathbf{r}}}{dt}$$

Velocity in terms of components

$$\vec{\mathbf{v}}(t) = v_x(t)\hat{\mathbf{i}} + v_y(t)\hat{\mathbf{j}} + v_z(t)\hat{\mathbf{k}}$$

Velocity components

$$v_x(t) = \frac{dx(t)}{dt} \quad v_y(t) = \frac{dy(t)}{dt} \quad v_z(t) = \frac{dz(t)}{dt}$$

Average velocity

$$\vec{\mathbf{v}}_{\text{avg}} = \frac{\vec{\mathbf{r}}(t_2) - \vec{\mathbf{r}}(t_1)}{t_2 - t_1}$$

Instantaneous acceleration

$$\vec{\mathbf{a}}(t) = \lim_{\Delta t \rightarrow 0} \frac{\vec{\mathbf{v}}(t+\Delta t) - \vec{\mathbf{v}}(t)}{\Delta t} = \frac{d\vec{\mathbf{v}}(t)}{dt}$$

Instantaneous acceleration, component form

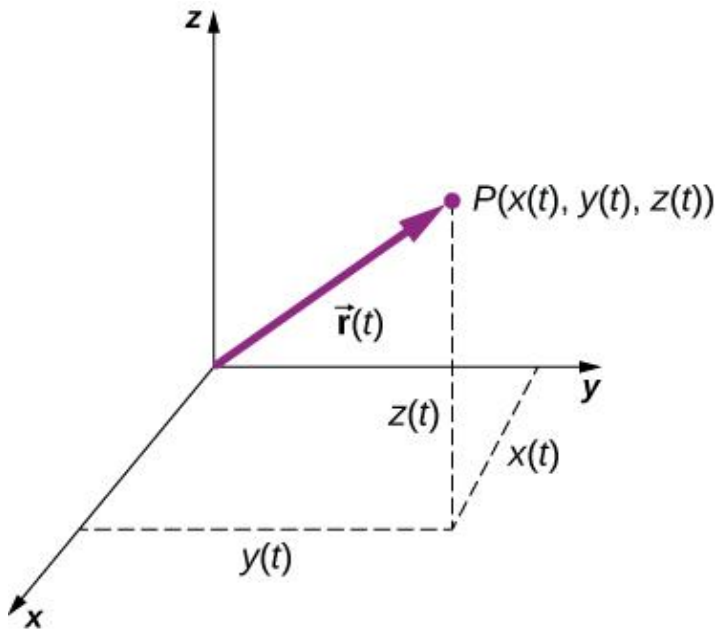
$$\vec{\mathbf{a}}(t) = \frac{dv_x(t)}{dt}\hat{\mathbf{i}} + \frac{dv_y(t)}{dt}\hat{\mathbf{j}} + \frac{dv_z(t)}{dt}\hat{\mathbf{k}}$$

Instantaneous acceleration as second derivatives of position

$$\vec{\mathbf{a}}(t) = \frac{d^2x(t)}{dt^2}\hat{\mathbf{i}} + \frac{d^2y(t)}{dt^2}\hat{\mathbf{j}} + \frac{d^2z(t)}{dt^2}\hat{\mathbf{k}}$$

4.1 Displacement and Velocity Vectors

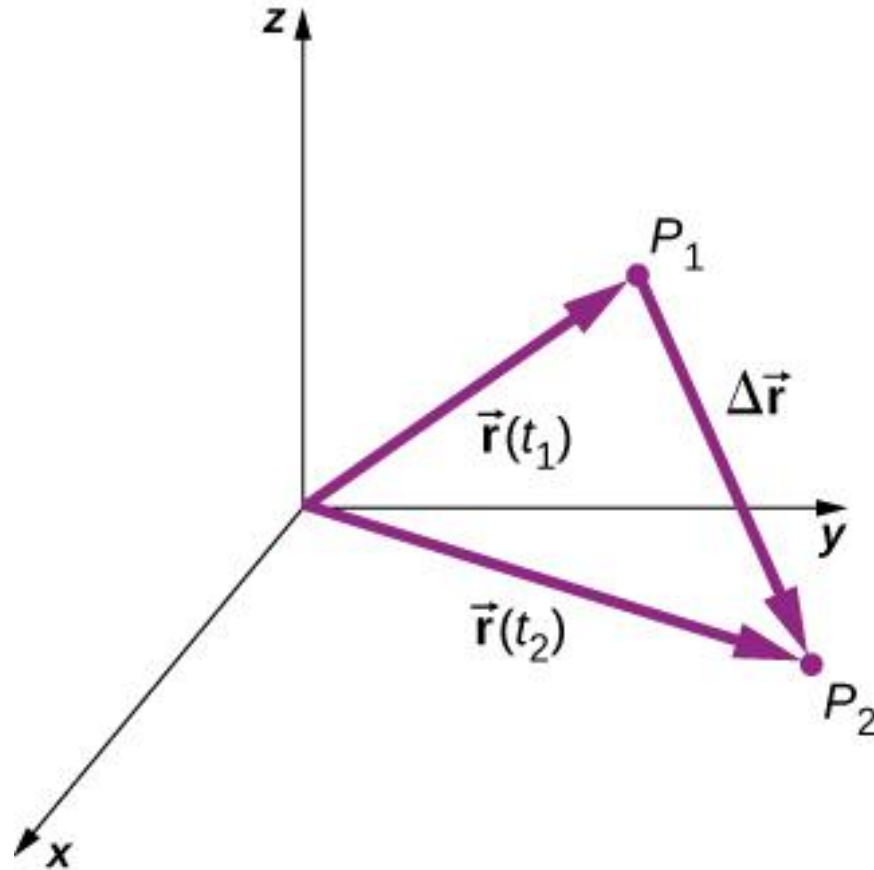
- 2D kinematics are similar to 1D kinematics.
- Must now use **full vector** notation rather than positive and negative signs to indicate direction of motion.
- **Position vector $\vec{\mathbf{r}}$** : drawn from origin of the coordinate system to the location of the particle.



$$\vec{\mathbf{r}}(t) = x(t)\hat{\mathbf{i}} + y(t)\hat{\mathbf{j}} + z(t)\hat{\mathbf{k}}.$$

A three-dimensional coordinate system with a particle at position $P(x(t), y(t), z(t))$.

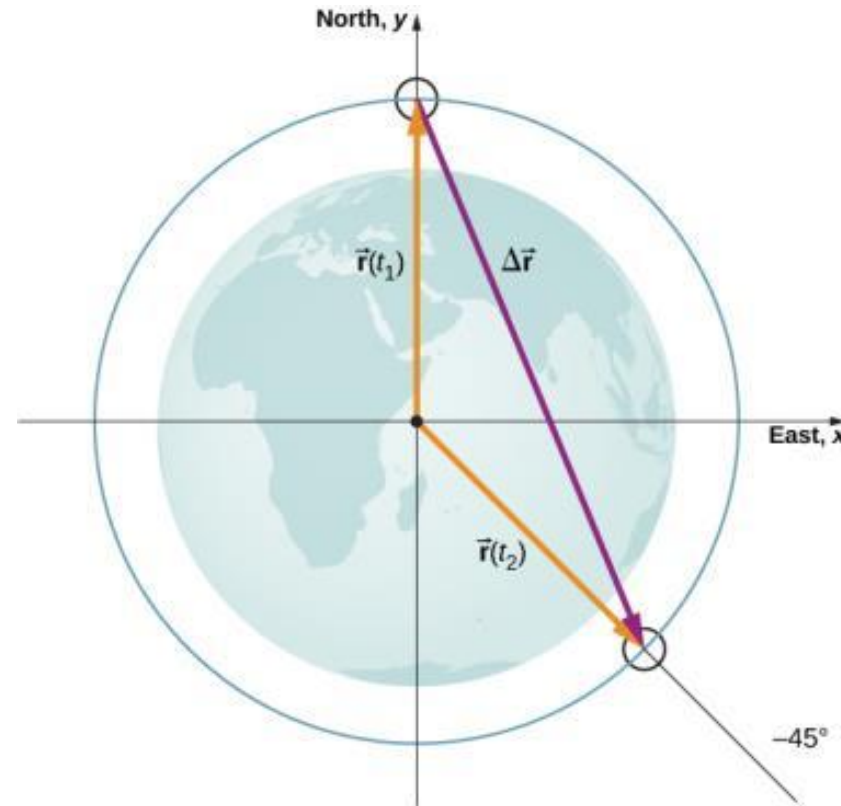
Displacement Vectors



The **displacement** $\Delta \vec{r} = \vec{r}(t_2) - \vec{r}(t_1)$ is the vector from P_1 to P_2 .

Displacement Vectors (Example 4.1)

A satellite is in a circular polar orbit around Earth at an altitude of 400 km, meaning it passes directly overhead at the North and South Poles. What is the magnitude and direction of the displacement vector from when it is directly over the North Pole to when it is at -45° latitude. (**Radius of Earth** = 6370 km)



Displacement Vectors (Example 4.1)

$$\vec{r}(t_1) = 6770. \text{ km} \hat{j}$$

$$\vec{r}(t_2) = 6770. \text{ km} (\cos(-45^\circ))\hat{i} + 6770. \text{ km} (\sin(-45^\circ))\hat{j}$$

Evaluating the sine and cosine, we have

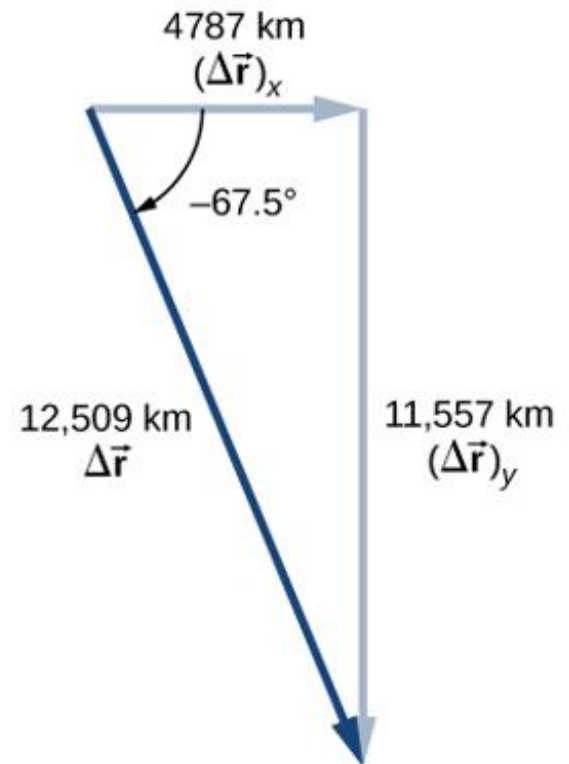
$$\vec{r}(t_1) = 6770. \hat{j}$$

$$\vec{r}(t_2) = 4787 \hat{i} - 4787 \hat{j}.$$

Now we can find $\Delta \vec{r}$, the displacement of the satellite:

$$\Delta \vec{r} = \vec{r}(t_2) - \vec{r}(t_1) = 4787 \hat{i} - 11,557 \hat{j}.$$

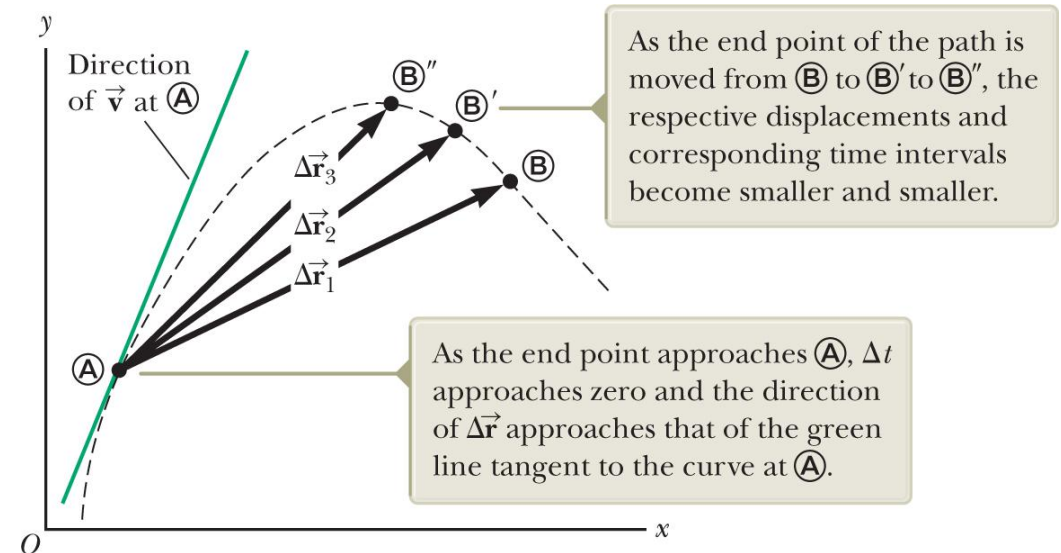
The magnitude of the displacement is $|\Delta \vec{r}| = \sqrt{(4787)^2 + (-11,557)^2} = 12,509 \text{ km}$. The angle the displacement makes with the x-axis is $\theta = \tan^{-1} \left(\frac{-11,557}{4787} \right) = -67.5^\circ$.



Average Velocity

- **Average velocity** of particle during time interval $\Delta t \rightarrow$ displacement of particle divided by time interval.

$$\vec{v}_{\text{avg}} = \frac{\vec{r}(t_2) - \vec{r}(t_1)}{t_2 - t_1}.$$

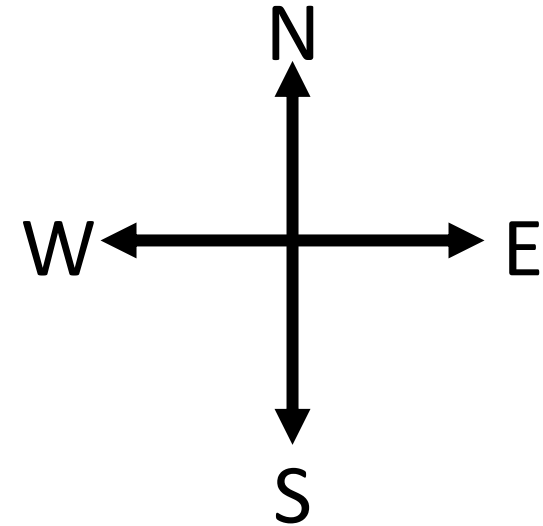


- Direction of average velocity vector is along $\Delta \vec{r}$.
- If particle starts motion at some point and returns to this point via any path: average velocity = 0 because displacement = 0.

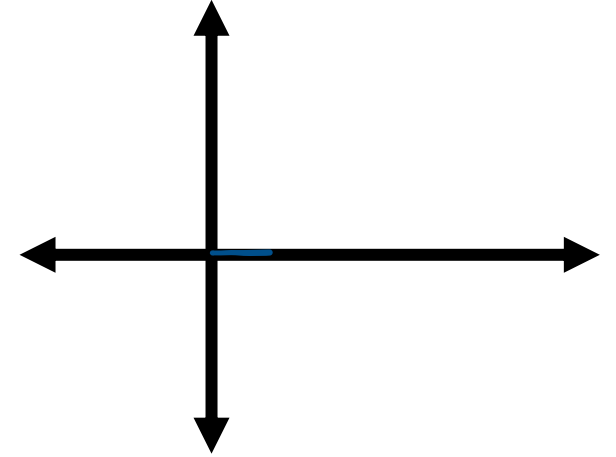
Problem solving (Textbook #20)

A bird flies straight 45 degrees above **East** a distance of 95.0 km for 3.0 hours. With the x -axis due east and the y -axis due north, what is the displacement in unit vector notation for the bird? What is the magnitude of the average velocity for the trip?

$$\vec{V}_{\text{avg}} = \frac{\vec{r}(t_2) - \vec{r}(t_1)}{t_2 - t_1} = \frac{\Delta \vec{r}}{\Delta t}$$

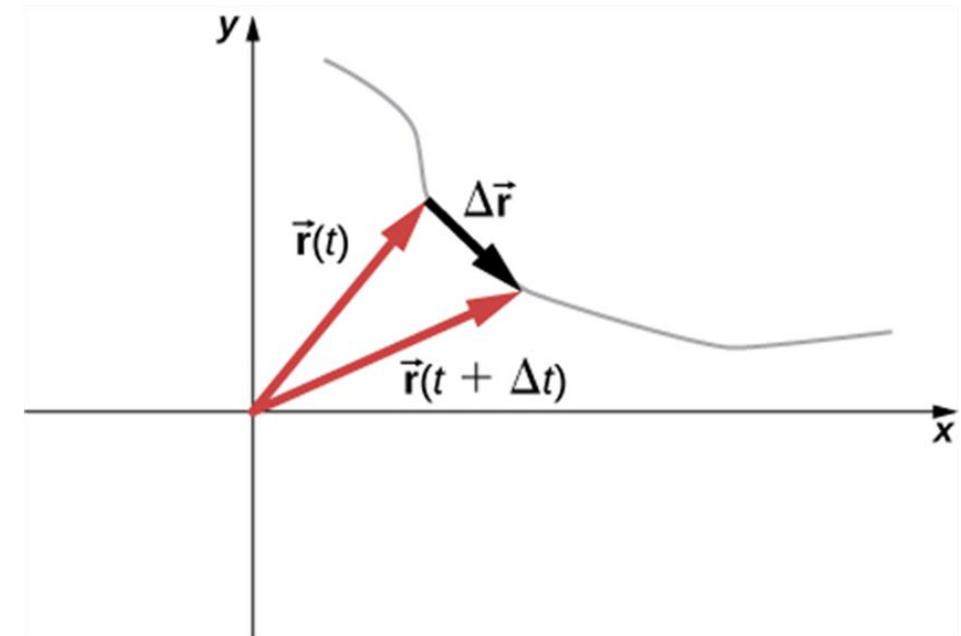
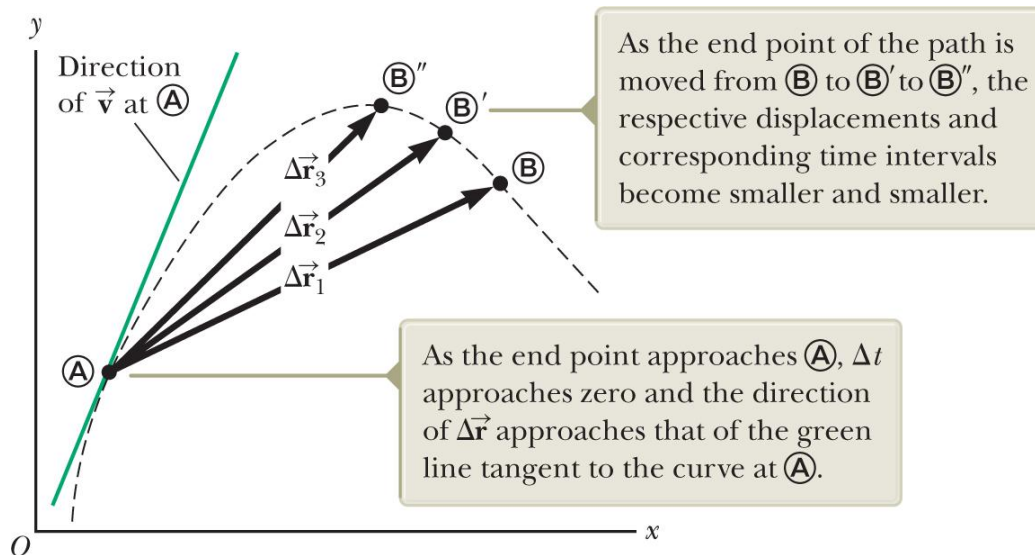


Problem solving (Textbook #20)



Instantaneous Velocity

- **Instantaneous velocity:** $\vec{v}(t) = \lim_{\Delta t \rightarrow 0} \frac{\vec{r}(t + \Delta t) - \vec{r}(t)}{\Delta t} = \frac{d\vec{r}}{dt}$. 4.4
- Direction of instantaneous velocity vector at any point is along line tangent to the path at that point.
- **Instantaneous speed:** $v = |\vec{v}|$



Velocity Vector Components

[Equation 4.4](#) can also be written in terms of the components of $\vec{\mathbf{v}}(t)$. Since

$$\vec{\mathbf{r}}(t) = x(t)\hat{\mathbf{i}} + y(t)\hat{\mathbf{j}} + z(t)\hat{\mathbf{k}},$$

we can write

$$\vec{\mathbf{v}}(t) = v_x(t)\hat{\mathbf{i}} + v_y(t)\hat{\mathbf{j}} + v_z(t)\hat{\mathbf{k}}$$

4.5

where

$$v_x(t) = \frac{dx(t)}{dt}, \quad v_y(t) = \frac{dy(t)}{dt}, \quad v_z(t) = \frac{dz(t)}{dt}.$$

4.6

Calculating Velocity Vector (Example 4.3)

The position function of a particle is, $\vec{r}(t) = [2.0 t^2 \hat{i} + (2.0 + 3.0 t) \hat{j} + 5.0 t \hat{k}] \text{ m}$

(a) What is the instantaneous velocity and speed at $t = 2.0 \text{ s}$?

(b) What is the average velocity between 1.0 s and 3.0 s ?

$$(a) \ v(t) = \frac{d\vec{r}(t)}{dt} = 4.0t\hat{i} + 3.0\hat{j} + 5.0\hat{k} \text{ m/s}$$

$$\vec{v}(2.0\text{s}) = 8.0\hat{i} + 3.0\hat{j} + 5.0\hat{k} \text{ m/s}$$

$$\text{Speed } |\vec{v}(2.0 \text{ s})| = \sqrt{8^2 + 3^2 + 5^2} = 9.9 \text{ m/s.}$$

(b) From [Equation 4.7](#),

$$\begin{aligned} \vec{v}_{\text{avg}} &= \frac{\vec{r}(t_2) - \vec{r}(t_1)}{t_2 - t_1} = \frac{\vec{r}(3.0 \text{ s}) - \vec{r}(1.0 \text{ s})}{3.0 \text{ s} - 1.0 \text{ s}} = \frac{(18\hat{i} + 11\hat{j} + 15\hat{k}) \text{ m} - (2\hat{i} + 5\hat{j} + 5\hat{k}) \text{ m}}{2.0 \text{ s}} \\ &= \frac{(16\hat{i} + 6\hat{j} + 10\hat{k}) \text{ m}}{2.0 \text{ s}} = 8.0\hat{i} + 3.0\hat{j} + 5.0\hat{k} \text{ m/s.} \end{aligned}$$

4.2 Acceleration Vectors

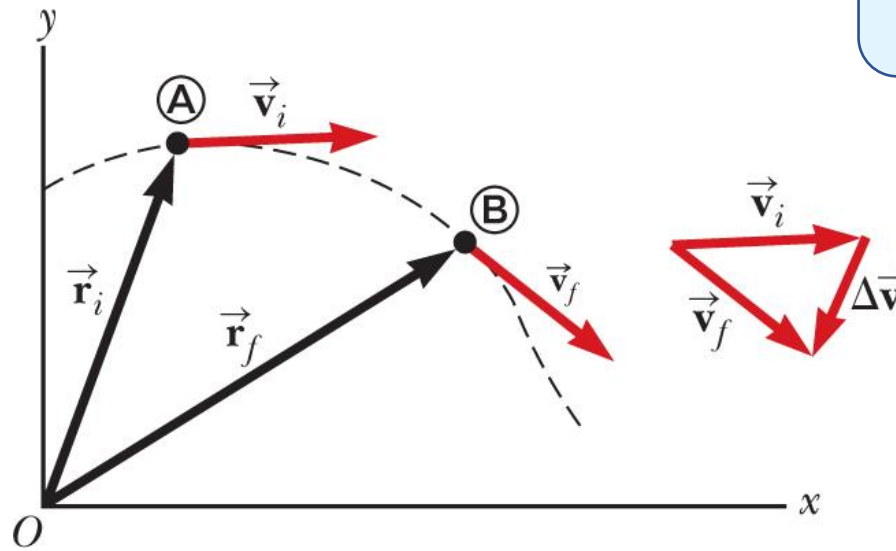
* Just for information how to calculate average acceleration if we know initial velocity at time (t_1) and final velocity at time (t_2).

Average acceleration:

$$\vec{a}_{\text{avg}} \equiv \frac{\Delta \vec{v}}{\Delta t} = \frac{\vec{v}_f - \vec{v}_i}{t_2 - t_1}$$

Instantaneous acceleration:

$$\vec{a} \equiv \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{v}}{\Delta t} = \frac{d\vec{v}}{dt}$$



Acceleration Vector Components

$$\vec{\mathbf{a}}(t) = \lim_{t \rightarrow 0} \frac{\vec{\mathbf{v}}(t + \Delta t) - \vec{\mathbf{v}}(t)}{\Delta t} = \frac{d\vec{\mathbf{v}}(t)}{dt}. \quad 4.8$$

The acceleration in terms of components is

$$\vec{\mathbf{a}}(t) = \frac{dv_x(t)}{dt} \hat{\mathbf{i}} + \frac{dv_y(t)}{dt} \hat{\mathbf{j}} + \frac{dv_z(t)}{dt} \hat{\mathbf{k}}. \quad 4.9$$

Also, since the velocity is the derivative of the position function, we can write the acceleration in terms of the second derivative of the position function:

$$\vec{\mathbf{a}}(t) = \frac{d^2 x(t)}{dt^2} \hat{\mathbf{i}} + \frac{d^2 y(t)}{dt^2} \hat{\mathbf{j}} + \frac{d^2 z(t)}{dt^2} \hat{\mathbf{k}}. \quad 4.10$$

Finding Acceleration Vector (Example 4.4)

If a particle has a velocity of $[\vec{v}(t) = 5.0 t \hat{i} + t^2 \hat{j} - 2.0 t^3 \hat{k}] \text{ m/s}$

(a) What is the acceleration function?

(b) What is the acceleration vector at 2.0 s? Find its magnitude and direction.

$$\begin{aligned} \text{(a)} \quad \vec{a}(t) &= \frac{d\vec{v}(t)}{dt} \\ &= (5.0 \hat{i} + 2.0 t \hat{j} - 6.0 t^2 \hat{k}) \text{ m/s}^2 \end{aligned}$$

$$\text{(b)} \quad \vec{a}(t = 2.0 \text{ s}) = (5.0 \hat{i} + 4.0 \hat{j} - 24.0 \hat{k}) \text{ m/s}^2$$

direction given in unit vector notation

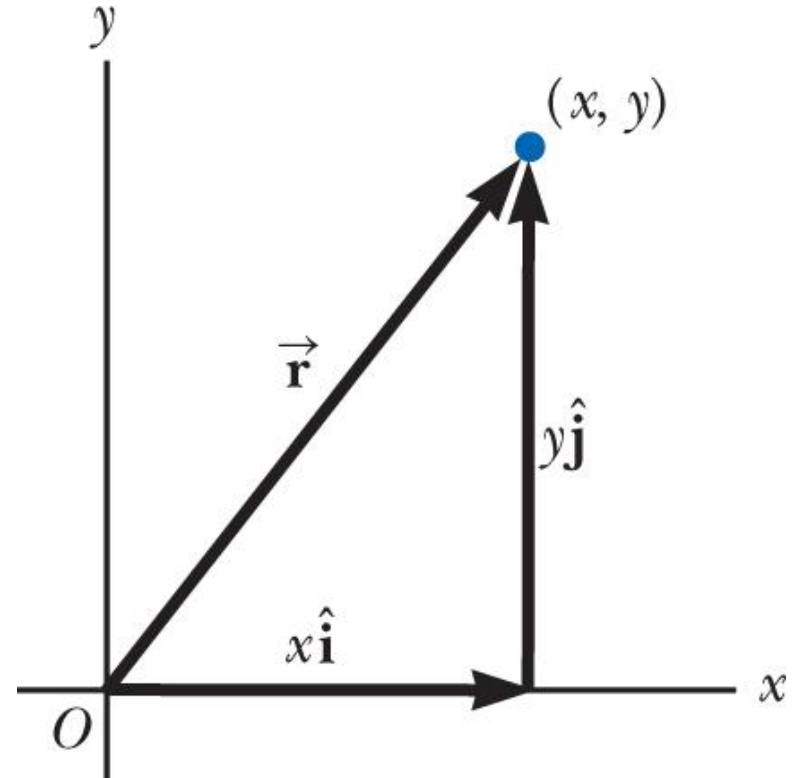
$$\begin{aligned} |\vec{a}(t = 2.0 \text{ s})| &= \sqrt{(5.0^2 + 4.0^2 + (-24.0)^2)} \text{ m/s}^2 \\ &= 24.8 \text{ m/s}^2 \text{ (magnitude)} \end{aligned}$$

4.3 Motion with Constant Acceleration in 2D

- Consider particle located in xy plane at position (x, y) .
- Any point can be specified by position vector $\mathbf{r}$ in unit-vector form:

$$\vec{\mathbf{r}} = x\hat{\mathbf{i}} + y\hat{\mathbf{j}}$$

- $\hat{\mathbf{i}}$ and $\hat{\mathbf{j}}$ are unit vectors.
- x , y , and $\mathbf{r}$ change with time as the particle moves.



Motion with Constant Acceleration in 2D

- Motions along **perpendicular** axes are independent and thus can be analyzed separately.
- Let us consider a 2D problem of a particle moving in the xy plane with constant acceleration. The acceleration vector is:

$$\vec{a} = a_{0x}\hat{i} + a_{0y}\hat{j}.$$

- Equations for position and velocity in the x and y -directions:

$$x(t) = x_0 + (v_x)_{\text{avg}}t$$

$$y(t) = y_0 + (v_y)_{\text{avg}}t$$

$$v_x(t) = v_{0x} + a_x t$$

$$v_y(t) = v_{0y} + a_y t$$

$$x(t) = x_0 + v_{0x}t + \frac{1}{2}a_x t^2$$

$$y(t) = y_0 + v_{0y}t + \frac{1}{2}a_y t^2$$

$$v_x^2(t) = v_{0x}^2 + 2a_x(x - x_0)$$

$$v_y^2(t) = v_{0y}^2 + 2a_y(y - y_0).$$

Motion with Constant Acceleration in 2D

- If position vector of particle is known, the velocity is:

$$\vec{v} = \frac{d\vec{r}}{dt} = \frac{dx}{dt}\vec{i} + \frac{dy}{dt}\vec{j} = v_x\vec{i} + v_y\vec{j}$$

- Each component of velocity can be expressed as:

$$v_x = v_{0x} + a_x t \text{ and } v_y = v_{0y} + a_y t$$

$$\vec{v} = (v_{0x} + a_x t)\vec{i} + (v_{0y} + a_y t)\vec{j} = (v_{0x}\vec{i} + v_{0y}\vec{j}) + (a_x\vec{i} + a_y\vec{j})t$$

$$\vec{v} = \vec{v}_0 + \vec{a}t \text{ (for constant } \vec{a}\text{)}$$

Motion with Constant Acceleration in 2D

- **x and y coordinates of particle under constant acceleration:**

$$x = x_0 + v_{0x}t + \frac{1}{2}a_xt^2 \quad y = y_0 + v_{0y}t + \frac{1}{2}a_yt^2$$

$$\vec{r} = \left(x_0 + v_{0x}t + \frac{1}{2}a_xt^2 \right) \vec{i} + \left(y_0 + v_{0y}t + \frac{1}{2}a_yt^2 \right) \vec{j}$$

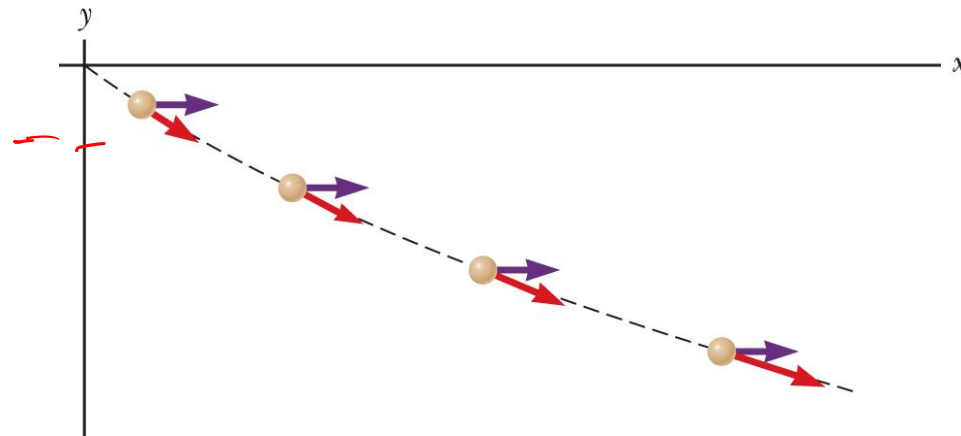
$$= (x_0\vec{i} + y_0\vec{j}) + (v_{0x}\vec{i} + v_{0y}\vec{j})t + \frac{1}{2}(a_x\vec{i} + a_y\vec{j})t^2$$

$$\vec{r} = \vec{r}_0 + \vec{v}_0t + \frac{1}{2}\vec{a}t^2 \quad (\text{for constant } \vec{a})$$

Example#1: Motion in a plane

A particle moves in the xy plane, starting from the origin at $t = 0$ with an initial velocity having an x -component of 20 m/s and a y -component of -15 m/s. The particle experiences an acceleration in the x direction, given by $a_x = 4.0$ m/s².

(A) Determine the velocity vector at any later time.



Example#1: Motion in a plane

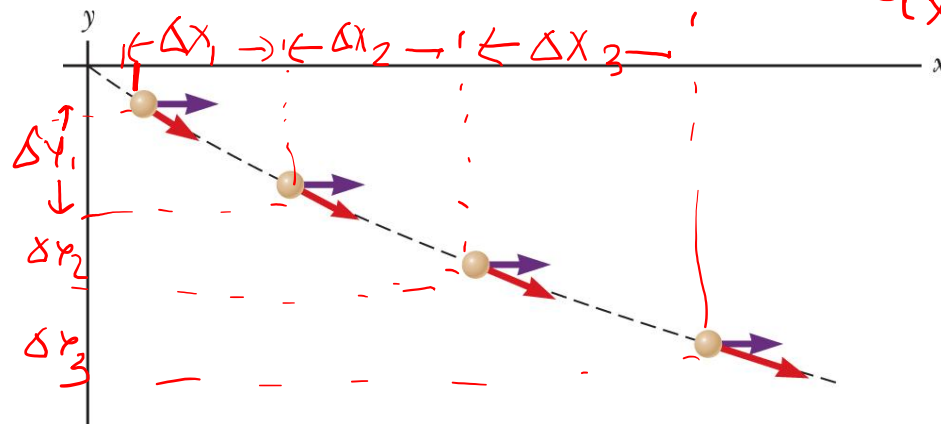
$$v_{0x} = 20 \text{ m/s}, v_{0y} = -15 \text{ m/s}, a_x = 4.0 \text{ m/s}, a_y = 0$$

$$\vec{v} = \vec{v}_0 + \vec{a}t = (v_{0x} + a_x t)\vec{i} + (v_{0y} + a_y t)\vec{j}$$

$$\vec{v} = [20 + (4.0)t]\vec{i} + [-15 + (0)t]\vec{j}$$

$$\vec{v} = [20 + (4.0)t]\vec{i} - 15\vec{j}$$

$a_y = 0$
 $\vec{\Delta y}_1 = \vec{\Delta y}_2 = \vec{\Delta y}_3$



$\vec{\Delta x}_3 > \vec{\Delta x}_2 > \vec{\Delta x}_1$

$\vec{a}_x \rightarrow$

Example#1: Motion in a plane

(B) Calculate the velocity and speed of the particle at $t = 5.0$ s and the angle the velocity vector makes with the x axis.

$$\vec{v} = [(20 + 4.0(5.0))\vec{i} - 15\vec{j}] = \boxed{(40\vec{i} - 15\vec{j}) \text{ m/s}}$$

$$v = |\vec{v}| = \sqrt{v_x^2 + v_y^2} = \sqrt{(40)^2 + (-15)^2} = \boxed{43 \text{ m/s}}$$

$$\theta = \tan^{-1} \left(\frac{v_y}{v_x} \right) = \tan^{-1} \left(\frac{-15 \text{ m/s}}{40 \text{ m/s}} \right) = -21^\circ$$

Example#1: Motion in a plane

(C) Determine the x and y coordinates of the particle at any time t and its position vector at this time.

$$\vec{r} = \vec{r}_0 + \vec{v}_0 t + \frac{1}{2} \vec{a} t^2$$

$$= \left(r_{0x} + v_{0x} t + \frac{1}{2} a_x t^2 \right) \vec{i} + \left(r_{0y} + v_{0y} t + \frac{1}{2} a_y t^2 \right) \vec{j}$$

$$\vec{r} = \left[0 + (20)t + \frac{1}{2} (4.0)t^2 \right] \vec{i} + \left[0 + (-15)t + \frac{1}{2} (0)t^2 \right] \vec{j}$$

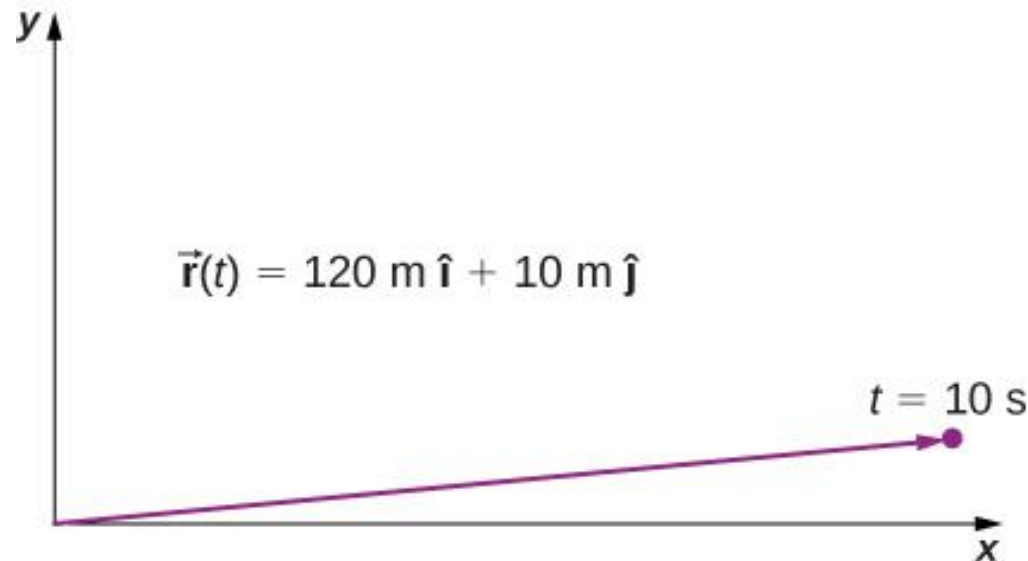
$$\vec{r} = \boxed{(20t + 2.0t^2)\vec{i} - 15t\vec{j}}$$

Problem solving (Textbook #28)

A boat leaves the dock at $t = 0$ and heads out into a lake with an **acceleration** of $2.0 \text{ m/s}^2 \hat{i}$. A strong wind is pushing the boat, giving it an **initial velocity** of $2.0 \text{ m/s} \hat{i} + 1.0 \text{ m/s} \hat{j}$

- (a) What is the velocity of the boat at $t = 10 \text{ s}$?
- (b) What is the position of the boat at $t = 10 \text{ s}$?

Draw a sketch of the boat's trajectory and position at $t = 10 \text{ s}$, showing the x - and y -axes.



Problem solving (Textbook #28)

Given: $\vec{a} = 2.0 \hat{i} \text{ (m/s}^2\text{)}$ $\left\{ \begin{array}{l} \text{compare} \\ \text{with} \end{array} \right.$ $\vec{a} = a_{0x} \hat{i} + a_{0y} \hat{j} \rightarrow a_{0x} = 2.0$
 $\vec{v}_0 = 2.0 \hat{i} + 1.0 \hat{j} \text{ (m/s)}$ $\vec{v}_0 = v_{0x} \hat{i} + v_{0y} \hat{j} \rightarrow v_{0x} = 2.0 \text{ \& } v_{0y} = 1.0$

a) velocity of the boat,

$$\vec{v} = v_x \hat{i} + v_y \hat{j} \text{ ——— (1)}$$

From the kinematic equation

$$\left. \begin{array}{l} v_x = v_{0x} + a_{0x} t \\ \quad = 2.0 + 2.0 t \end{array} \right\} \begin{array}{l} v_y = v_{0y} + a_{0y} t \\ \quad = 1.0 + 0 \end{array} \text{ ——— (2)}$$

Equation (1) and (2) gives

$$\vec{v} = (2.0 + 2.0 t) \hat{i} + (1.0) \hat{j}$$

$$\vec{v}(t = 10s) = [2.0 + 2.0(10s)] \hat{i} + 1.0 \hat{j}$$

$$\boxed{\vec{v}(10s) = (22 \hat{i} + 1.0 \hat{j}) \text{ m/s}} \text{ ——— (3)}$$

Problem solving (Textbook #28)

b) The position of the boat $\vec{r}(t) = x\hat{i} + y\hat{j}$ — (4)

From the kinematic equation of position

$$x(t) = v_{0x}t + \frac{1}{2}a_{0x}t^2$$

$$x(10s) = (2.0)(10s) + \frac{1}{2}(2.0)(10s)^2$$

$$= 20 + 100$$

$$x(10s) = 120m \quad \text{--- (5)}$$

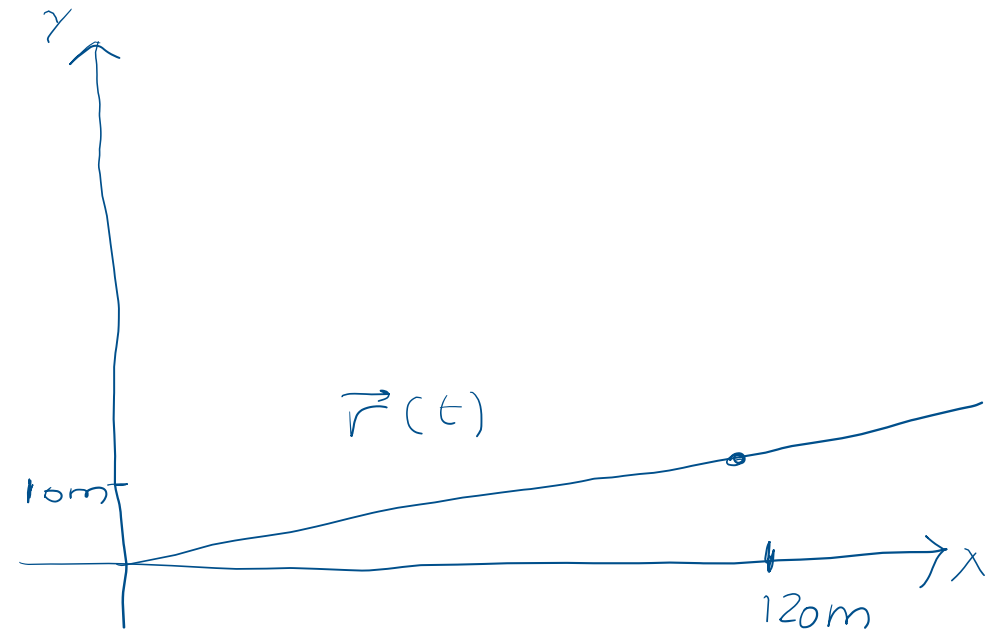
$$y(t) = v_{0y}t + \frac{1}{2}a_{0y}t^2$$

$$= (1.0)(10s) + \frac{1}{2}(0)(10)^2$$

$$= 10m \quad \text{--- (6)}$$

From equation (4) (5) and (6)

$$\vec{r}(t) = (120m)\hat{i} + (10m)\hat{j}$$



CQ#1

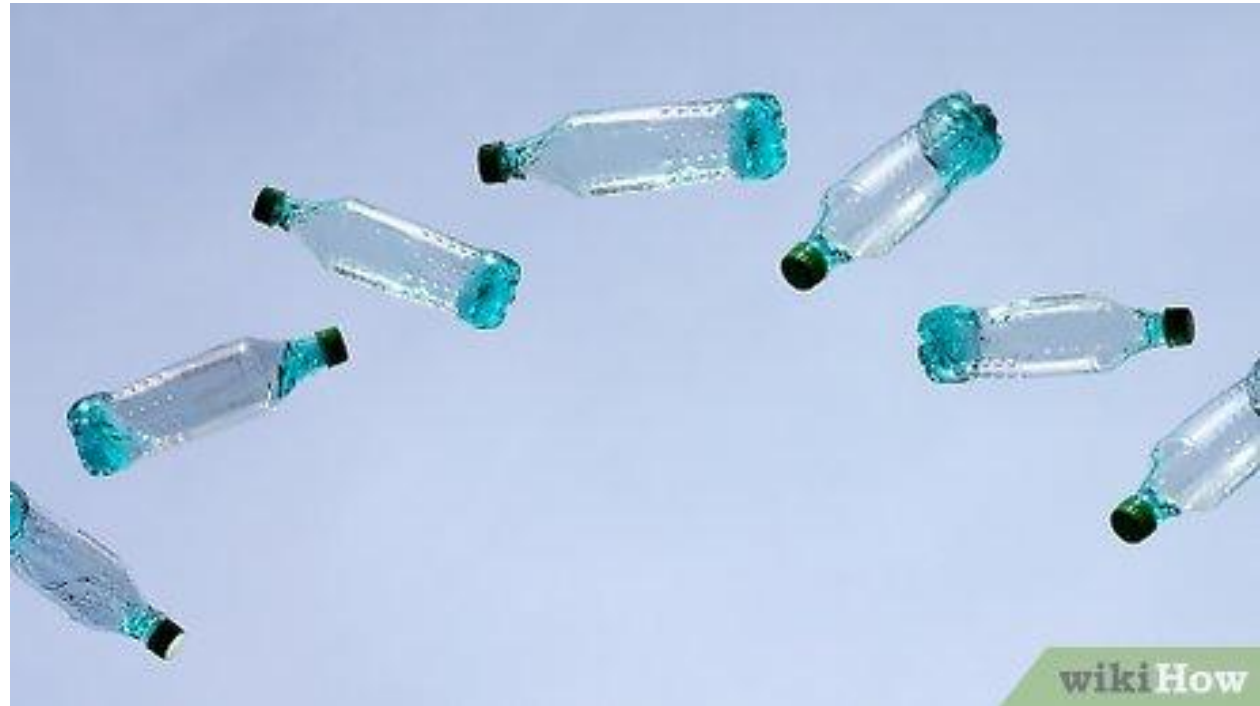
You toss your water bottle across the room. What is the **acceleration** in the y-direction while it is in the air?

a. -9.80 m/s^2

b. 0.00 m/s^2

c. 9.80 m/s^2

d. Not enough information given



4.4 Projectile Motion

- **Projectile motion** is the motion of an object thrown or projected into the air, **subject only to acceleration projectiles, gravity**.
- Such objects are called **projectiles**, and their path is called a **trajectory**.
- Path of projectile (its trajectory) is *always* parabola.
- Simple to analyze with two assumptions:
 1. Free-fall acceleration constant over range of motion and directed downward (i.e., $a_x = 0$, $a_y = -g$);
 2. Effect of air resistance is negligible.

(I) Projectile Motion (Projection angle θ_0)

$$V_{0x} = V_0 \cos \theta_0$$

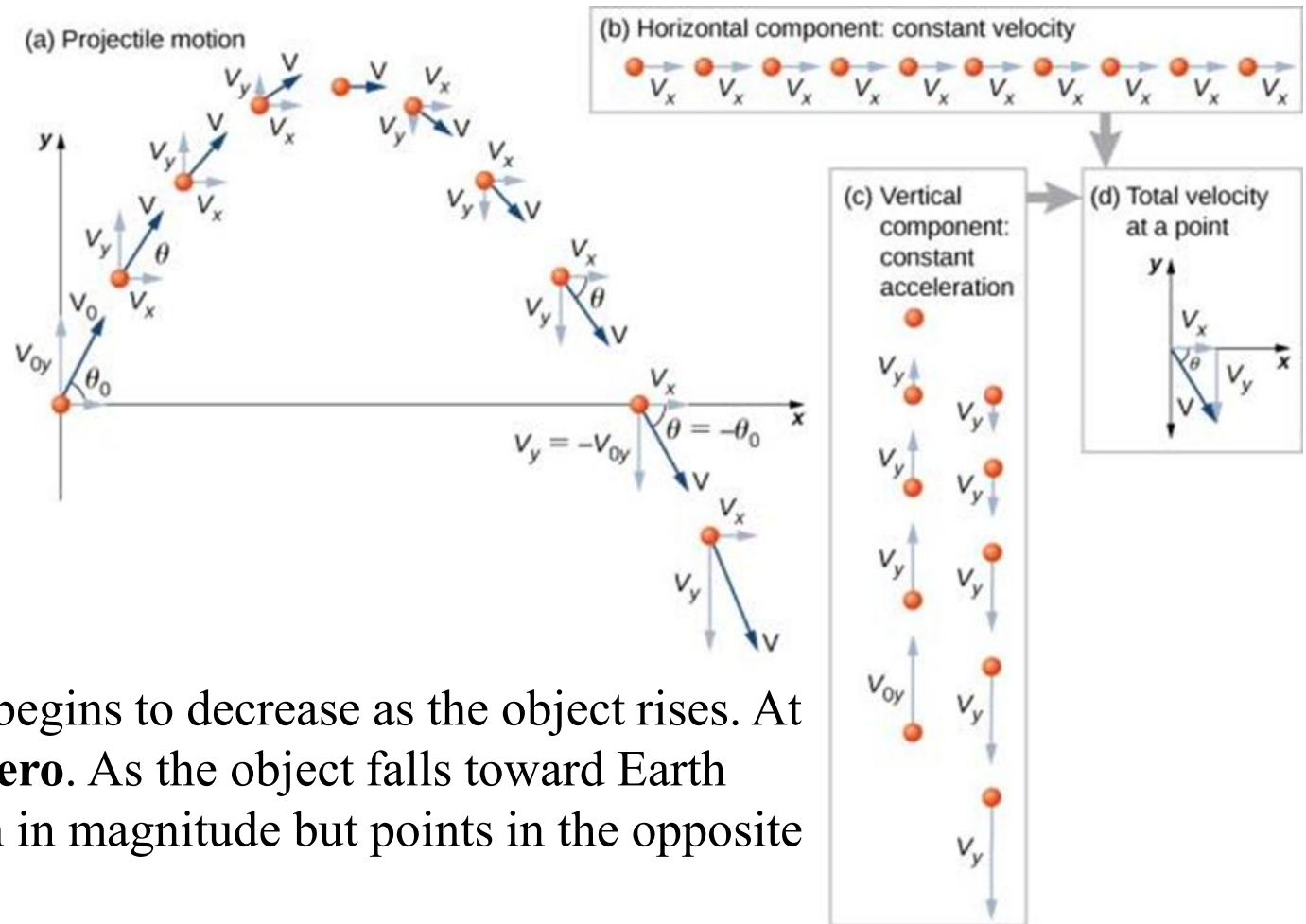
$$V_{0y} = V_0 \sin \theta_0$$

(a) We break 2D projectile motion into two independent 1D motions along the vertical and horizontal axes.

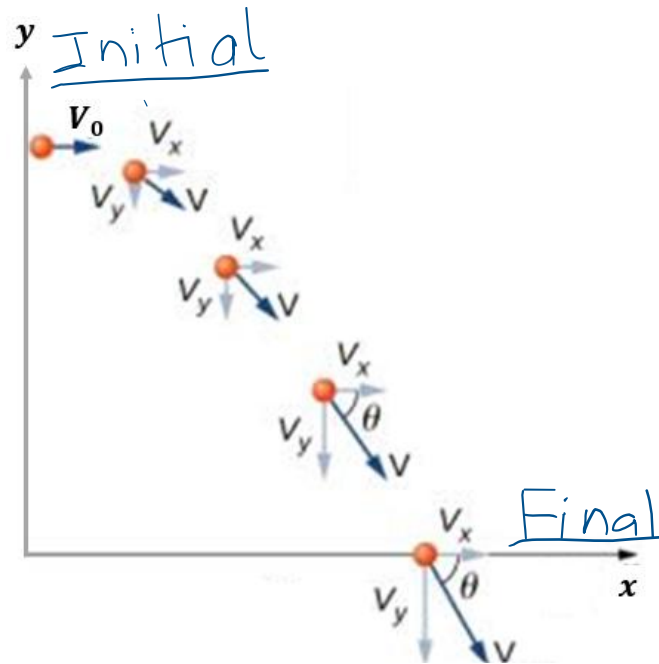
(b) The horizontal motion is simple, because $a_x = 0$ and v_x is a constant.

(c) The velocity in the vertical direction begins to decrease as the object rises. At its highest point, the vertical velocity is **zero**. As the object falls toward Earth again, the vertical velocity increases again in magnitude but points in the opposite direction to the initial vertical velocity.

(d) The x and y motions are recombined to give the **total velocity** at any given point on the trajectory.

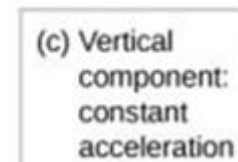
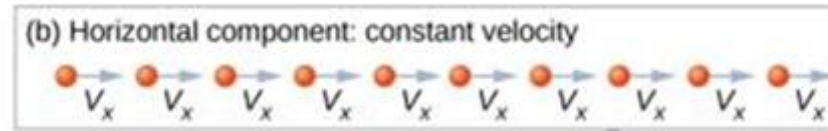


(II) Projectile Motion (Projection angle $\theta_0 = 0$)



$$V_{0x} = V_0 \cos \theta_0 \quad \boxed{V_{0x} = V_0}$$

$$V_{0y} = V_0 \sin \theta_0 \quad \boxed{V_{0y} = 0}$$



(d) Total velocity at a point

A vector diagram showing the total velocity V at a point. The horizontal component is V_x and the vertical component is V_y . The resultant vector V is at an angle θ below the horizontal. Handwritten blue notes provide the following equations:

$$\begin{aligned} V_x &= V_{0x} \\ V_y &= V_{0y} - gt \\ V_y &= -gt \end{aligned}$$

The magnitude of the total velocity is given by:

$$\text{Magnitude } |V| = \sqrt{(V_x)^2 + (V_y)^2}$$

Direction $\theta = \tan^{-1} \left(\frac{V_y}{V_x} \right)$

(I) Projectile Motion (Projection angle θ_0)

$$V_{0x} = V_0 \cos \theta_0$$

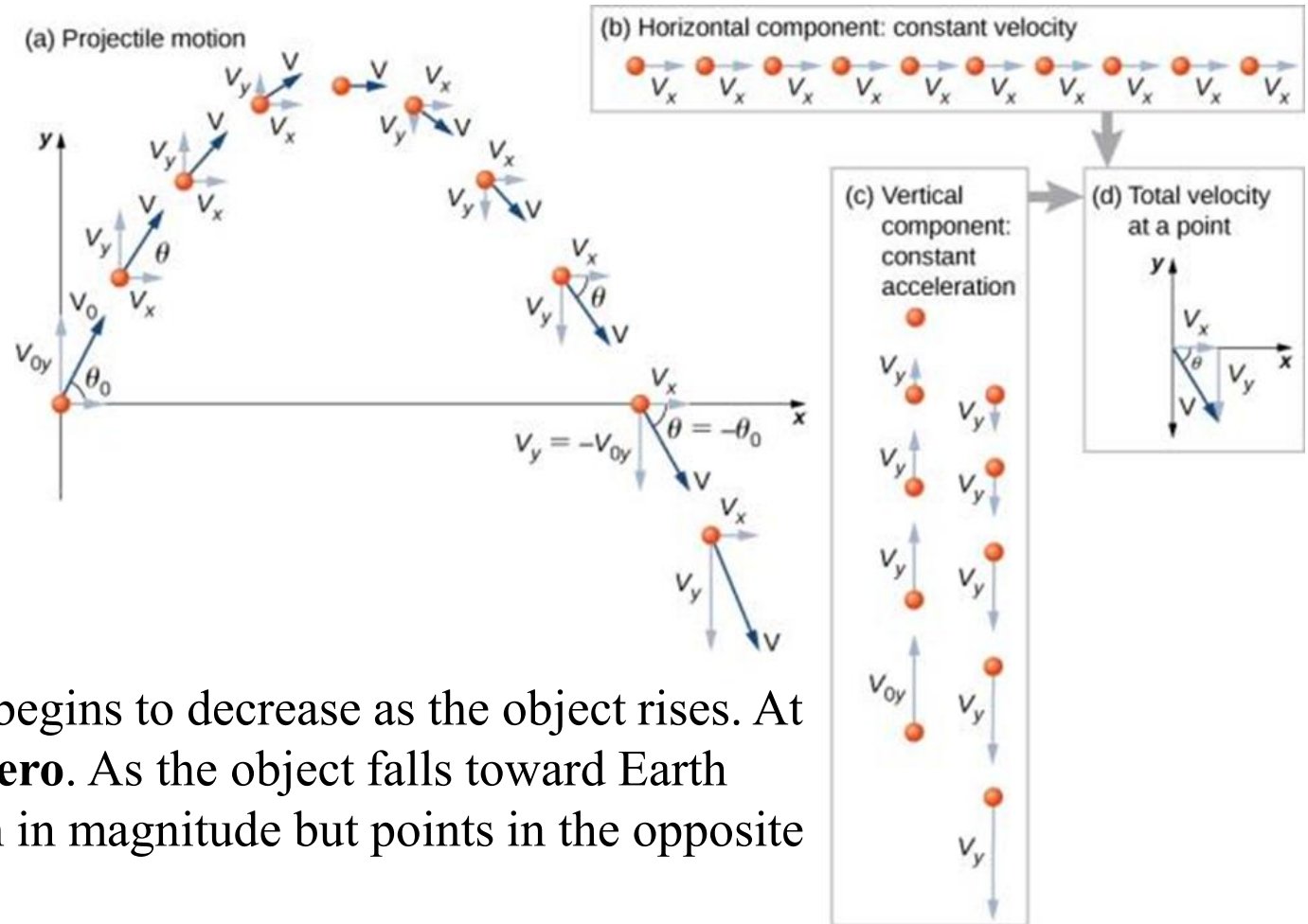
$$V_{0y} = V_0 \sin \theta_0$$

(a) We break 2D projectile motion into two independent 1D motions along the vertical and horizontal axes.

(b) The horizontal motion is simple, because $a_x = 0$ and v_x is a constant.

(c) The velocity in the vertical direction begins to decrease as the object rises. At its highest point, the vertical velocity is **zero**. As the object falls toward Earth again, the vertical velocity increases again in magnitude but points in the opposite direction to the initial vertical velocity.

(d) The x and y motions are recombined to give the **total velocity** at any given point on the trajectory.



(I)Projectile Motion (Projection angle θ_0)

- **Horizontal** and **vertical** components of projectile's motion are completely independent and can be handled separately:
- Horizontal direction: constant velocity motion ($a_x = 0$);
- Vertical direction: constant acceleration motion ($a_y = -g$).

Horizontal Motion

$$V_{0x} = V_0 \cos \theta_0$$

$$v_{0x} = v_x, \quad x = x_0 + v_x t \quad (4.19)$$

Vertical Motion

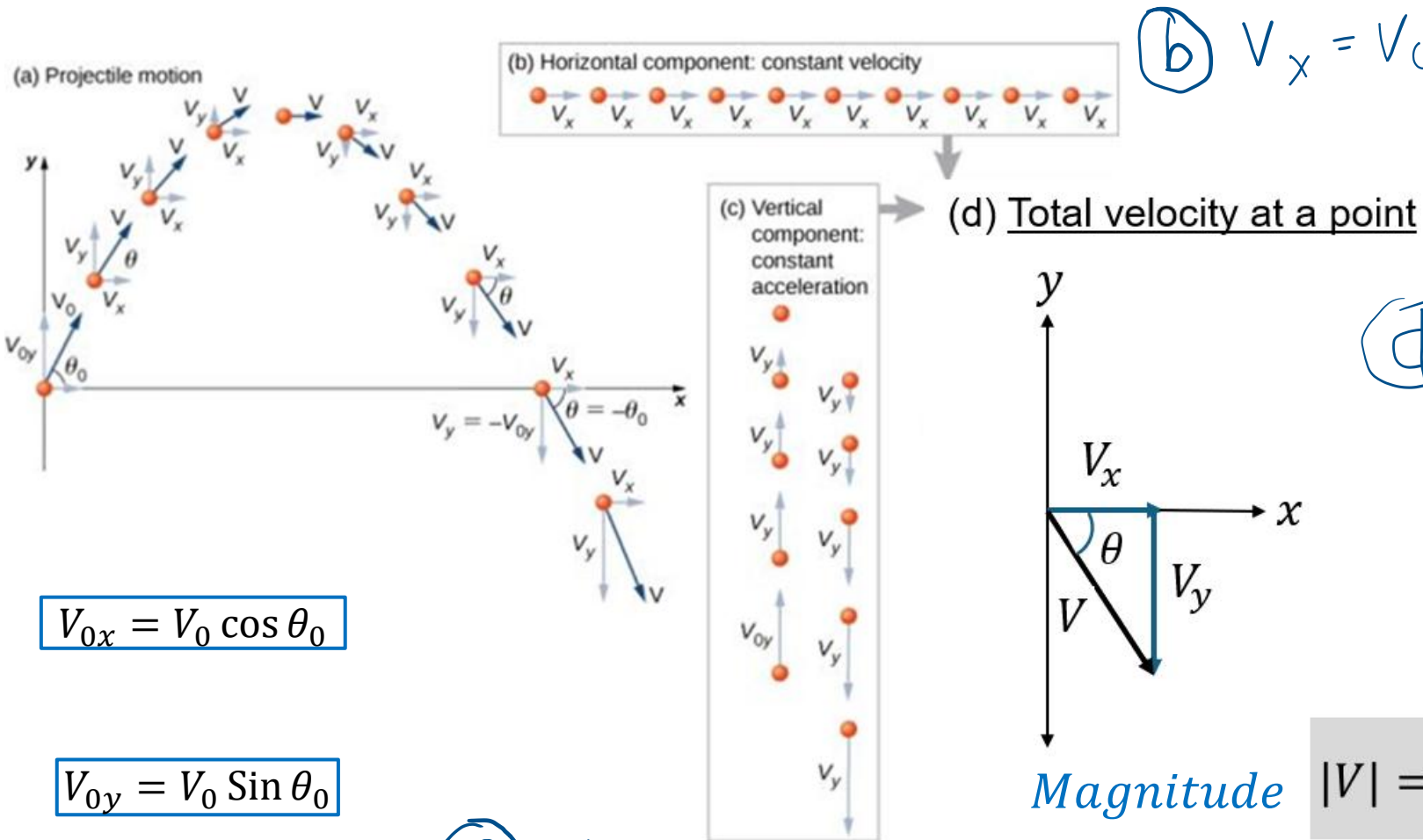
$$V_{0y} = V_0 \sin \theta_0$$

$$y = y_0 + \frac{1}{2}(v_{0y} + v_y)t$$
$$v_y = v_{0y} - gt \quad (4.21)$$

$$y = y_0 + v_{0y}t - \frac{1}{2}gt^2 \quad (4.22)$$

$$v_y^2 = v_{0y}^2 - 2g(y - y_0) \quad (4.23)$$

(I) Projectile Motion (Projection angle θ_0)



⑥ $V_x = V_{0x}$ [constant]

①

$$V_{0x} = V_0 \cos \theta_0$$

$$V_{0y} = V_0 \sin \theta_0$$

③ V_y (variable)

$V_y = 0$ [maximum height]

④ Final velocity

$$V_x = V_{0x}$$
$$V_y = V_{0y} - gt$$

Magnitude

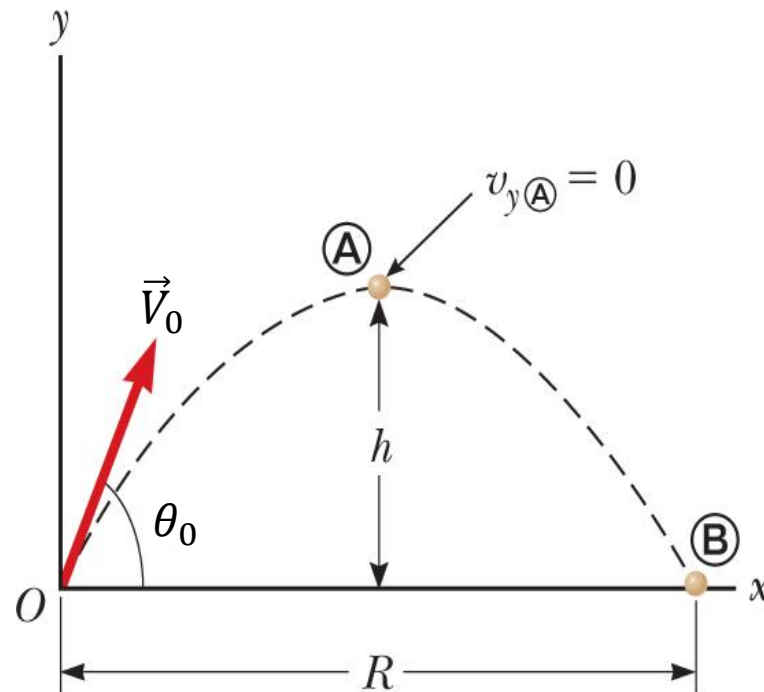
$$|V| = \sqrt{(V_x)^2 + (V_y)^2}$$

Direction

$$\theta = \tan^{-1} \left(\frac{V_y}{V_x} \right)$$

Projectile Motion (terms)

- **Trajectory** – Flight path of object
- **Time of Flight** (T_{toF}) – total time object spends in the air
- **Range** (R) – horizontal displacement of the object
- **Maximum Height** (h) – highest vertical position object reaches during flight



Height (h) of a Projectile

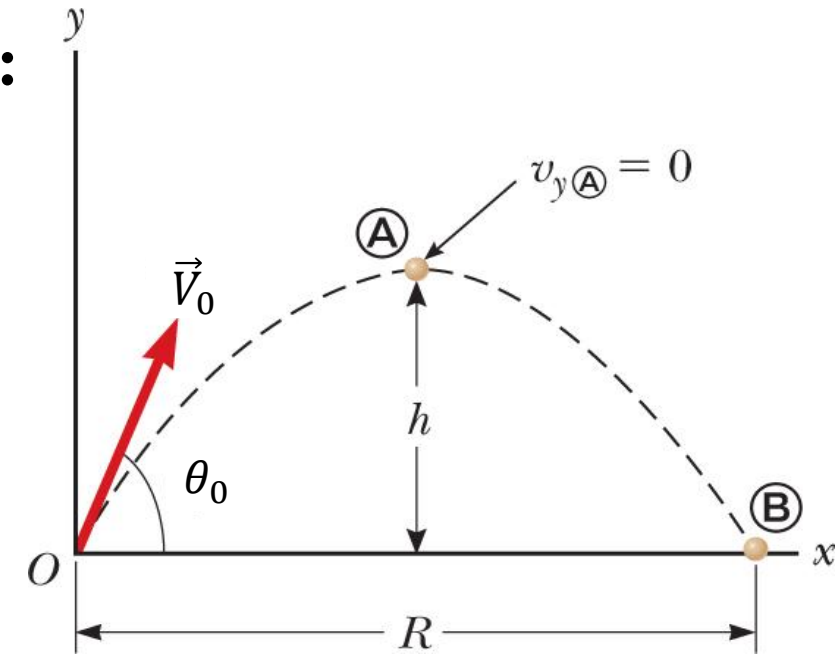
- Find h and R mathematically in terms of v_0 , θ_0 , and g :

$$v_y = v_{0y} - gt$$

$$\Rightarrow 0 = v_0 \sin \theta_0 - gt_A$$

$$\Rightarrow t_A = \frac{v_0 \sin \theta_0}{g}$$

$$T_{tof} = 2 t_A = \frac{2 v_0 \sin \theta_0}{g}$$



$$y = y_0 + v_{0y}t - \frac{1}{2}gt^2$$

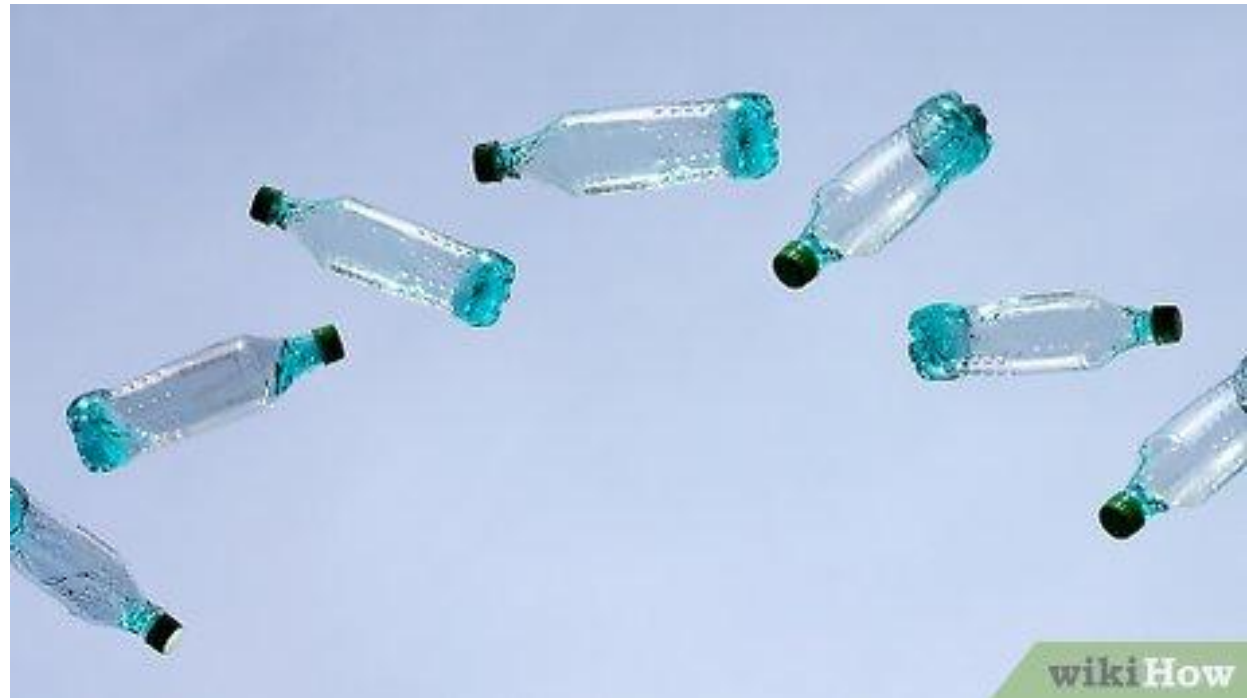
$$h = 0 + (v_0 \sin \theta_0) \frac{v_0 \sin \theta_0}{g} - \frac{1}{2}g \left(\frac{v_0 \sin \theta_0}{g} \right)^2$$

$$h = \frac{v_0^2 \sin^2 \theta_0}{2g}$$

CQ#2

I toss a ball upward from the ground. How does the **time** it takes to reach the top compare to the time it takes to fall from the top to the ground? (ignore air resistance!)

- a. Less
- b. More
- c. Same
- d. Can't tell



Horizontal Range (R) of a Projectile

$$x = x_0 + v_{0x}t$$

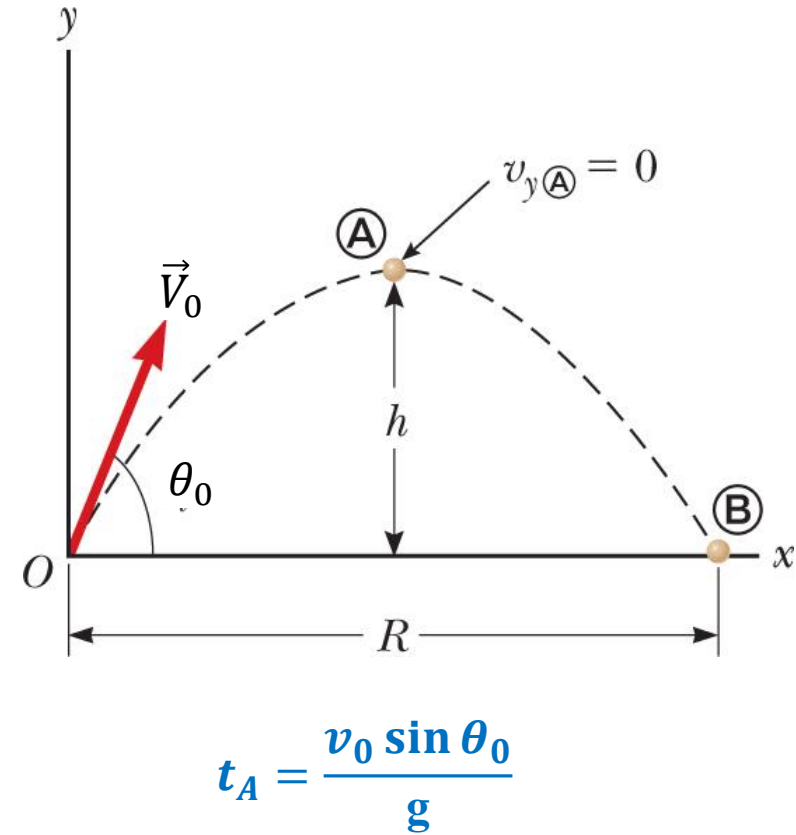
$$\Rightarrow R = v_{0x} t_B = (v_0 \cos \theta_0) 2t_A$$

$$= (v_0 \cos \theta_0) 2 \left(\frac{v_0 \sin \theta_0}{g} \right)$$

$$= \frac{2v_0^2 \sin \theta_0 \cos \theta_0}{g}$$

$$2 \sin \theta_0 \cos \theta_0 = \sin 2 \theta_0$$

$$R = \frac{v_0^2 \sin 2 \theta_0}{g}$$

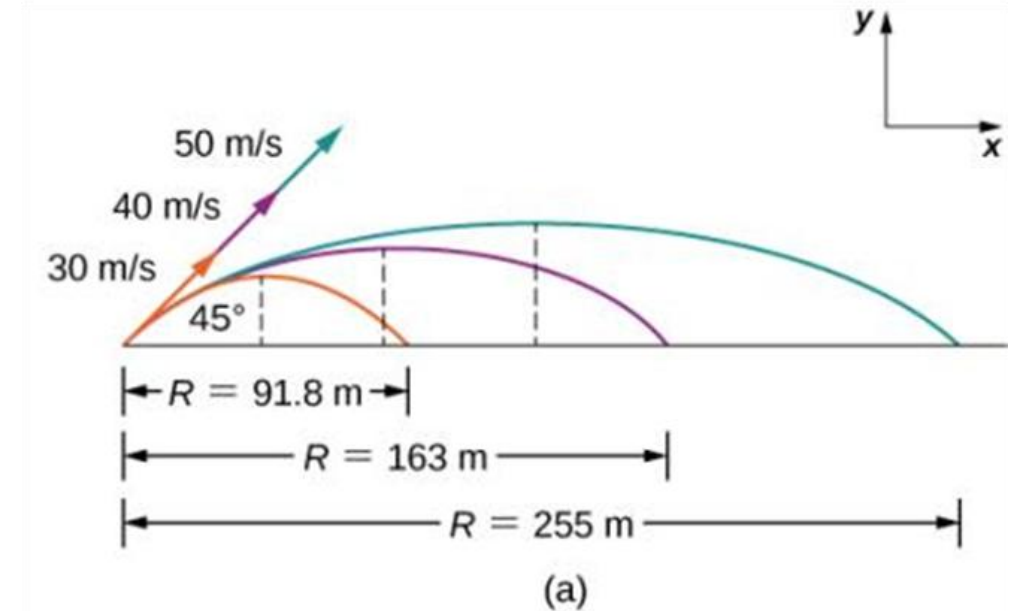


$$R_{max} = \frac{v_0^2}{g} \text{ when } \theta_0 = 45^\circ$$

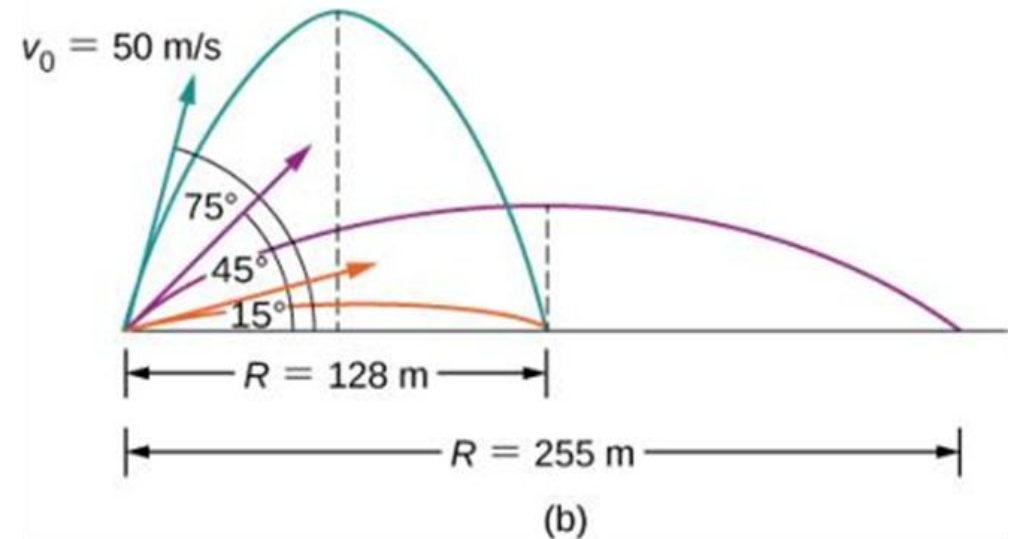
Trajectories of a Projectile

- **Trajectories of projectiles on level ground.**

(a) The greater the initial speed v_0 , the greater the range for a given initial angle.



(b) The effect of initial angle θ_0 on the range of a projectile with a given initial speed. Note that the range is the same for initial angles of 15° and 75°, although the maximum heights of those paths are different.



Example #2 (Range and maximum height)

A long jumper leaves the ground at an angle of 20.0° above the horizontal and at a speed of 11.0 m/s .

(A) How **far** does he jump in the horizontal direction?

$$\begin{aligned} R &= \frac{v_0^2 \sin 2\theta_0}{g} \\ &= \frac{(11.0 \text{ m/s})^2 \sin(2 \cdot 20.0^\circ)}{9.80 \text{ m/s}^2} \\ &= \boxed{7.94 \text{ m}} \end{aligned}$$



(B) What is the **maximum height** reached?

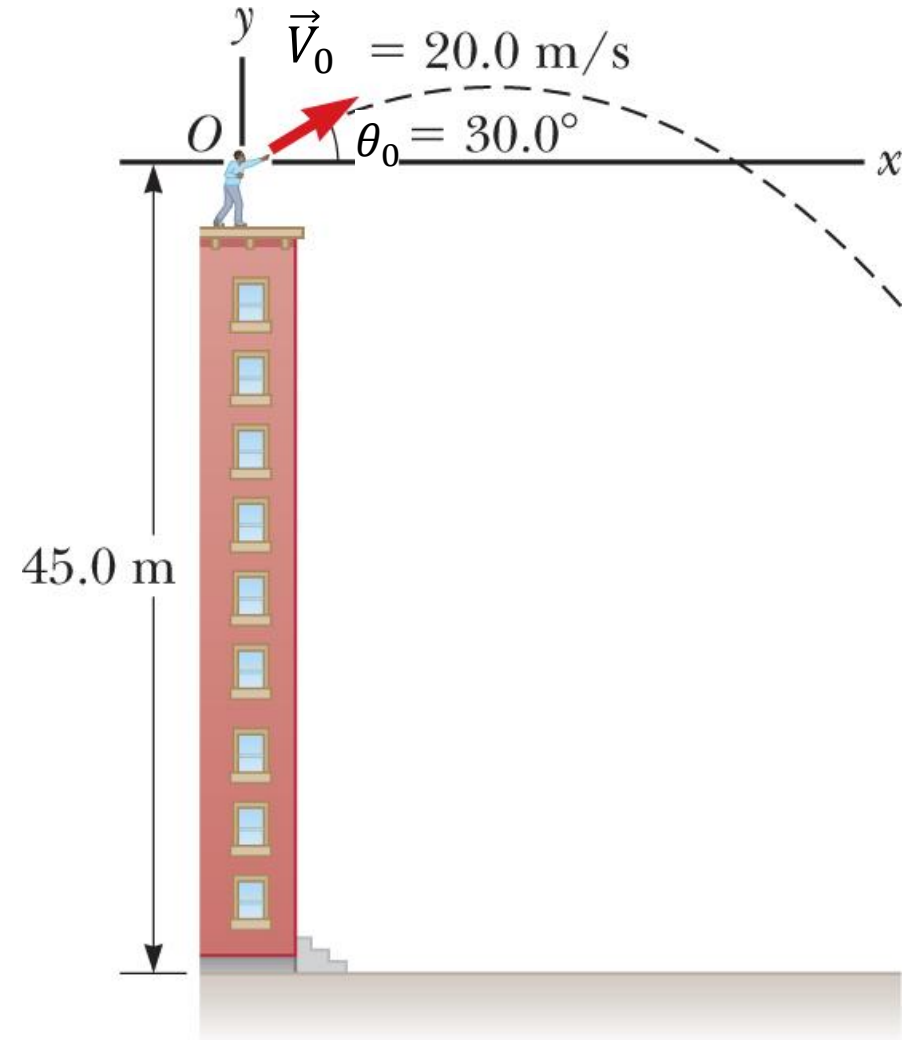
$$\begin{aligned} h &= \frac{v_0^2 \sin^2 \theta_0}{2g} = \frac{(11.0 \text{ m/s})^2 (\sin 20.0^\circ)^2}{2(9.80 \text{ m/s}^2)} \\ &= \boxed{0.722 \text{ m}} \end{aligned}$$

Example #3 (stone thrown at an angle)

A stone is thrown from the top of a building upward at an angle of 30.0° to the horizontal with an initial speed of 20.0 m/s as shown in the figure. The height from which the stone is thrown is 45.0 m above the ground.

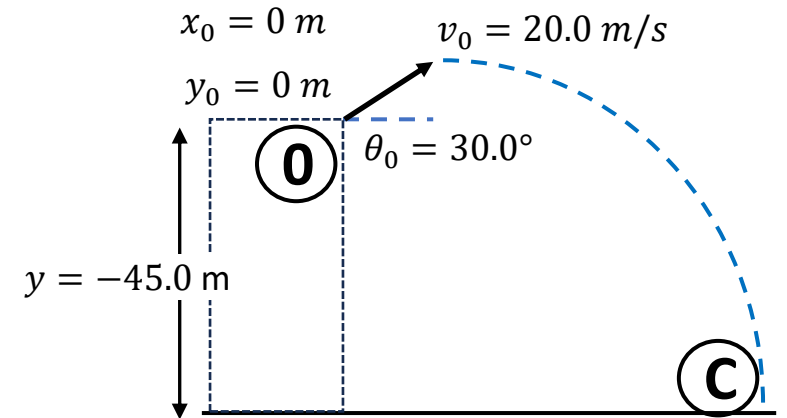
(A) **How long** does it take the stone to reach the ground?

(B) What is the **speed** of the stone just before it strikes the ground?



Example #3 (stone thrown at an angle)

Given, $x_0 = y_0 = 0$,
 $y = -45.0 \text{ m}$,
 $a_y = -g$,
 $v_0 = 20.0 \text{ m/s}$
 $\theta_0 = 30^\circ$



$$\begin{aligned}v_{0x} &= v_0 \cos \theta_0 \\&= (20.0 \text{ m/s}) \cos 30^\circ \\&= 17.3 \text{ m/s}\end{aligned}$$

$$\begin{aligned}v_{0y} &= v_0 \sin \theta_0 \\&= (20.0 \text{ m/s}) \sin 30^\circ \\&= 10.0 \text{ m/s}\end{aligned}$$

$$y = y_0 + v_{0y}t + \frac{1}{2}a_y t^2$$

$$-45.0 \text{ m} = 0 + (10.0 \text{ m/s})t + \frac{1}{2}(-9.80 \text{ m/s}^2)t^2$$

$$-45.0 = 10.0 t - 4.90 t^2$$

This is quadratic equation in “t”, find the roots and take only positive root because time cannot be negative!!

$$t = \boxed{4.22 \text{ s}}$$

Example #3 (stone thrown at an angle)

(B) What is the speed of the stone just before it strikes the ground?

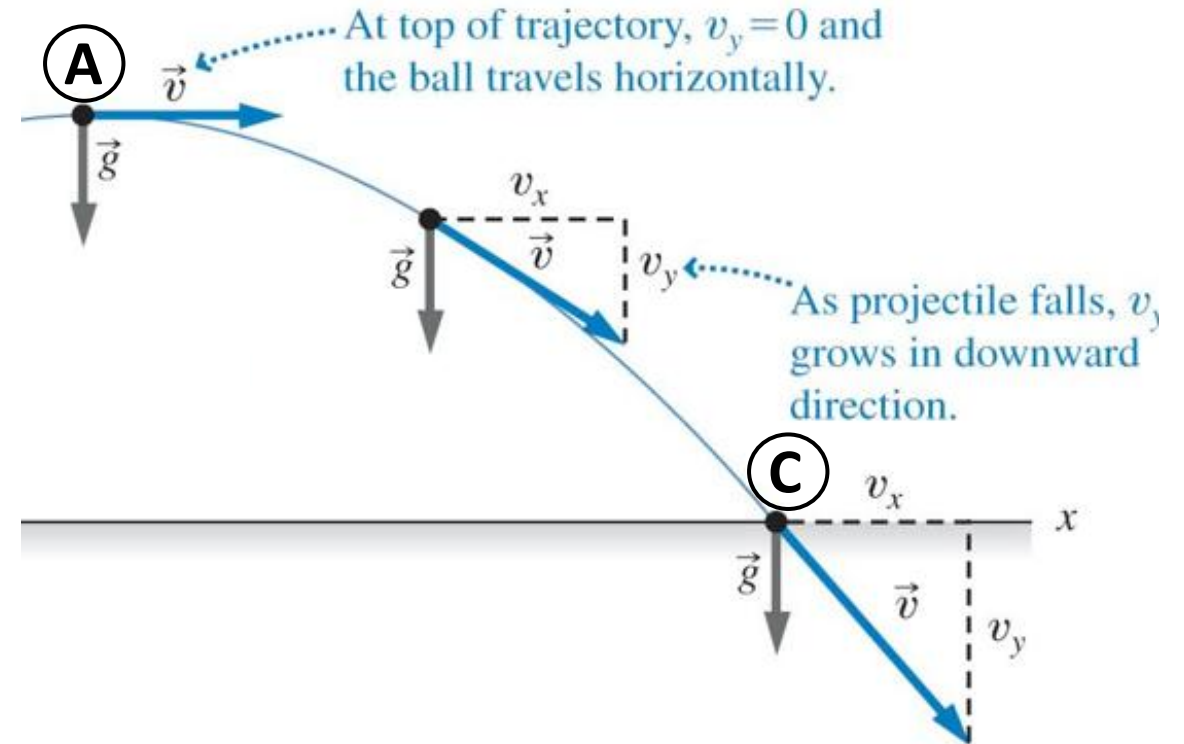
$$|\vec{v}| = \sqrt{(v_x)^2 + (v_y)^2}$$

$$v_x = v_{0x} = 17.3 \text{ m/s}$$

$$\begin{aligned} v_y &= v_{0y} + a_y t \\ &= 10.0 \text{ m/s} + (-9.80 \text{ m/s}^2)(4.22 \text{ s}) \\ &= 10.0 - 41.3 \end{aligned}$$

$$v_y = -31.3 \text{ m/s}$$

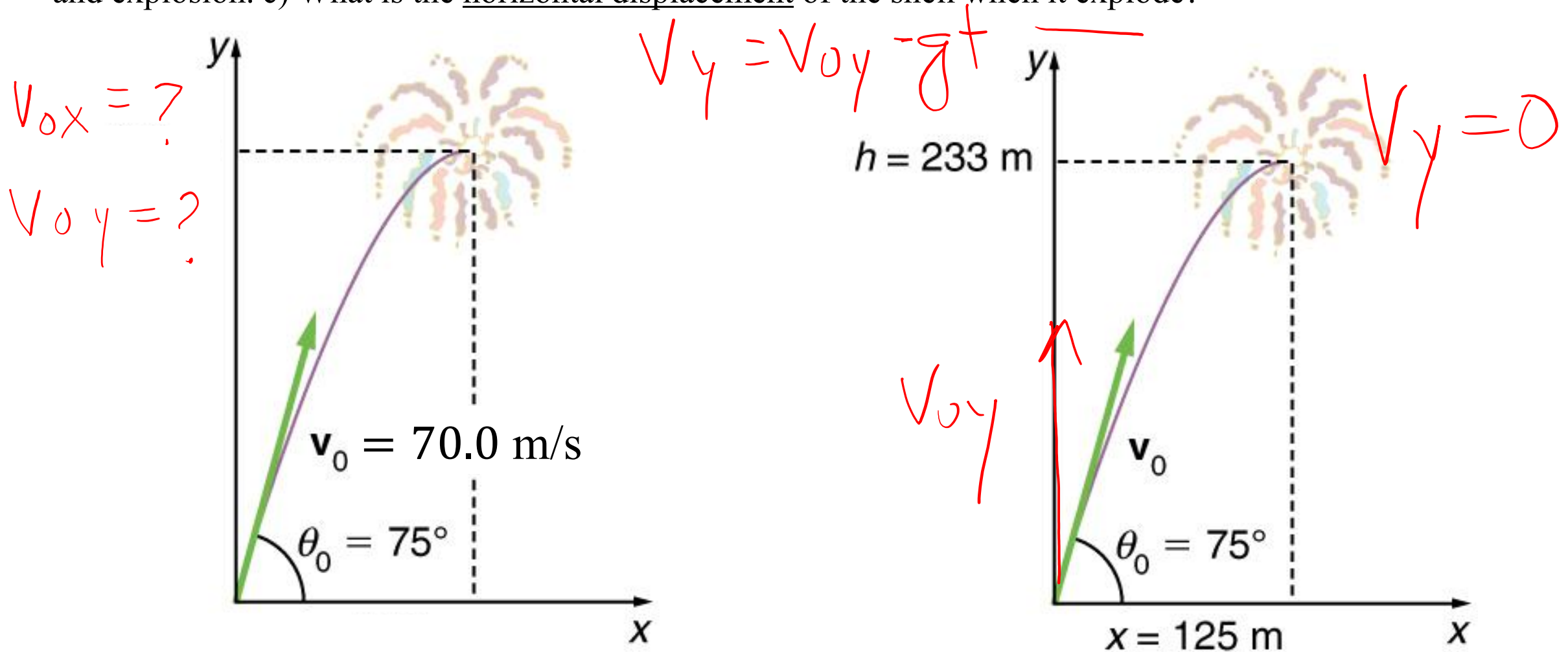
$$|\vec{v}| = \sqrt{(17.3)^2 + (-31.3)^2} = 35.8 \text{ m/s}$$



Example 4.7: Firework Projectile (Textbook)

During a fire work display, a shell is shot into the air with the initial speed of 70.0 m/s at an angle of 75.0 °.

- a) Calculate the height at which the shell explode. b) How much time passes between the launch of the shell and explosion. c) What is the horizontal displacement of the shell when it explode?



Example 4.7: Firework Projectile (Textbook)

$$\begin{array}{l|l}
 \begin{array}{l}
 v_0 = 70.0 \text{ m/s} \\
 \theta_0 = 75.0^\circ \\
 x_0 = 0, y_0 = 0 \\
 t = ? \quad y = ? \quad x = ?
 \end{array} & \begin{array}{l}
 \text{x-axis} \\
 v_{0x} = v_0 \cos \theta_0 \\
 x = x_0 + v_{0x} t \quad \text{--- (1)}
 \end{array} \\
 \hline
 & \begin{array}{l}
 \text{y-axis} \\
 v_{0y} = v_0 \sin \theta_0 \\
 v_y = v_{0y} - gt \quad \text{--- (2)} \\
 y = y_0 + v_{0y} t - \frac{1}{2} gt^2 \quad \text{--- (3)} \\
 v_y^2 = v_{0y}^2 - 2g(y - y_0) \quad \text{--- (4)}
 \end{array}
 \end{array}$$

a) Let 't' be time to reach maximum height.

Use (2) $v_y = v_{0y} - gt$
 $0 = v_0 \sin \theta_0 - gt \Rightarrow t = \frac{v_0 \sin \theta_0}{g} = 6.89 \text{ s}$

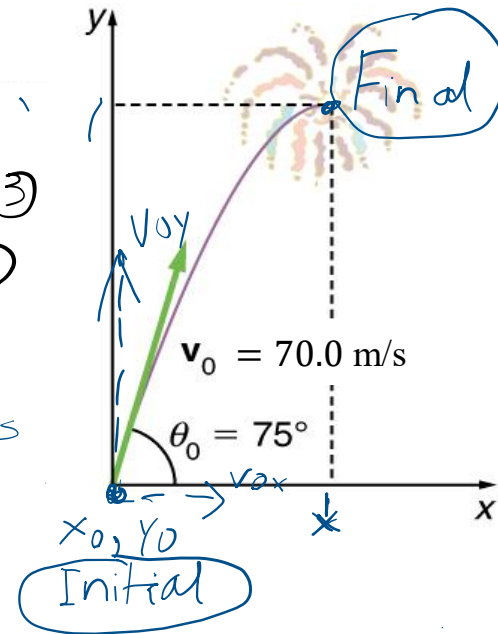
b) At time 't' the y-position or height is,

Use (3) $y = y_0 + v_{0y} t - \frac{1}{2} gt^2$
 OR $= 0 + v_0 \sin \theta_0 (6.89) - \frac{1}{2} (9.80) (6.89)^2 \Rightarrow \boxed{y = 233 \text{ m}}$
 Use (4)

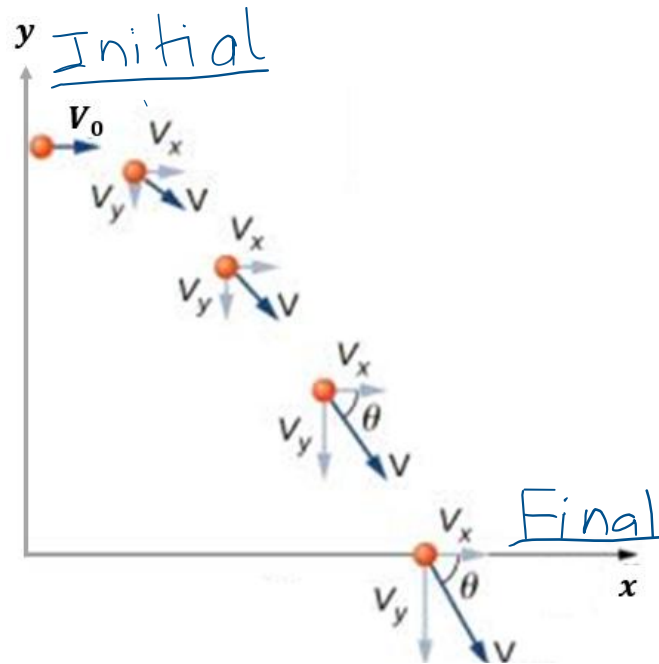
c) At time 't', the x-position or horizontal distance

Use (1) $x = x_0 + v_{0x} t$
 $= 0 + v_0 \cos \theta_0 (6.89)$

$\boxed{x = 125 \text{ m}}$

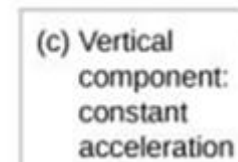
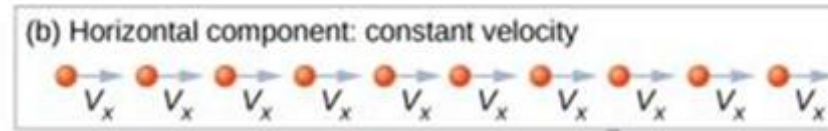


(II) Projectile Motion (Projection angle $\theta_0 = 0$)



$$V_{0x} = V_0 \cos \theta_0 \quad \boxed{V_{0x} = V_0}$$

$$V_{0y} = V_0 \sin \theta_0 \quad \boxed{V_{0y} = 0}$$



(d) Total velocity at a point

A vector diagram showing the total velocity V at a point. The horizontal component is V_x and the vertical component is V_y . The angle between V and the horizontal axis is θ . Handwritten blue notes provide the equations for the components: $V_x = V_{0x}$, $V_y = V_{0y} - gt$, and $V_y = -gt$. The magnitude of the total velocity is given by the equation: $|V| = \sqrt{(V_x)^2 + (V_y)^2}$.

Magnitude $|V| = \sqrt{(V_x)^2 + (V_y)^2}$

Direction $\theta = \tan^{-1} \left(\frac{V_y}{V_x} \right)$

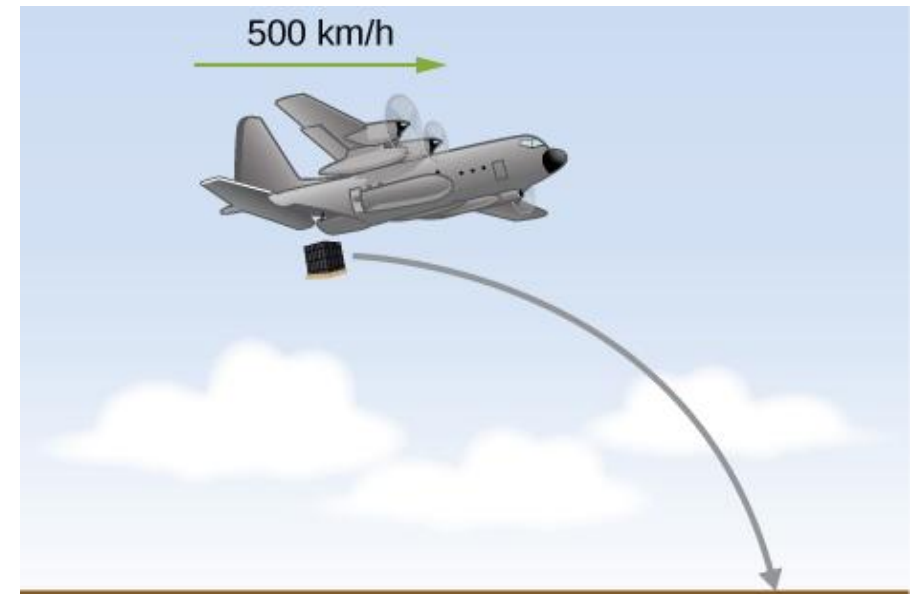
Problem solving (Textbook #36)

An airplane flying horizontally with a speed of 500 km/h at a height of 800 m drops a crate of supplies (see the following figure). If the parachute fails to open, how far in front of the release point does the crate hit the ground?

[Answer: 1779 m]

Note: The x-component of initial velocity $\vec{v}_{0x}$ is 500 km/h.

The y-component of initial velocity $\vec{v}_{0y}$ is ZERO.



Problem solving (Textbook #36)

$$\begin{aligned}
 V_{ox} &= 500 \text{ km/h} = \dots \text{ m/s} \\
 &= 500 \frac{\text{km}}{\text{h}} \times \frac{1000 \text{ m}}{1 \text{ km}} \times \frac{1 \text{ h}}{3600 \text{ s}} \\
 &= \dots \text{ m/s} \quad \text{--- (1)}
 \end{aligned}$$

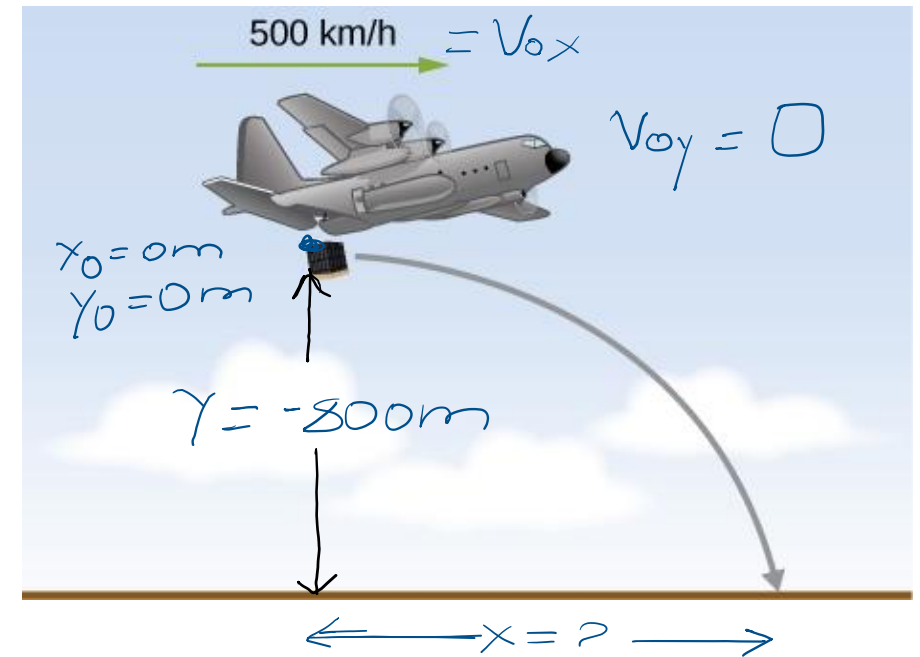
Now, let's use equation along y direction to find time for the crate to reach the ground.

$$\begin{aligned}
 y &= y_0 + V_{oy} t - \frac{1}{2} g t^2 \\
 -800 &= 0 + 0 - \frac{1}{2} (9.80) t^2 \\
 800 &= 4.90 t^2
 \end{aligned}$$

$$\frac{800}{4.90} = t^2$$

$$t = \sqrt{\frac{800}{4.90}}$$

$$\boxed{t = \dots \text{ s}} \quad \text{--- (2)}$$



Now, the same time can be used to find the distance along x direction,

$$x = x_0 + V_{ox} t$$

↑ from (1)
 ↑ from (2)

$$\boxed{x = \dots \text{ m}}$$

Example #4: Waymo

An autonomously driven car (**Waymo**) malfunctioned and drove off a 20.0 m high cliff, landing below at 30.0 m from the base of the cliff. Determine (a) the time it took for the car to reach the ground (b) the initial speed of the car (c) the final speed and direction of the car when it crashed.

(a) $t = 2.20 \text{ s}$

(b) $v_0 = v_{0x} = 14.8 \text{ m/s}$

(c) $v = 24.8 \text{ m/s}$ and $\theta = 15.3^\circ$



Example #4: Waymo

Initial

$$V_{0x} = V_0 = ?$$

$$V_{0y} = 0 \text{ (Projection angle } \theta_0 = 0)$$

$$Y_0 = 0 \text{ m, } Y = -20.0 \text{ m}$$

$$x_0 = 0 \text{ m, } x = 30.0 \text{ m}$$

Final

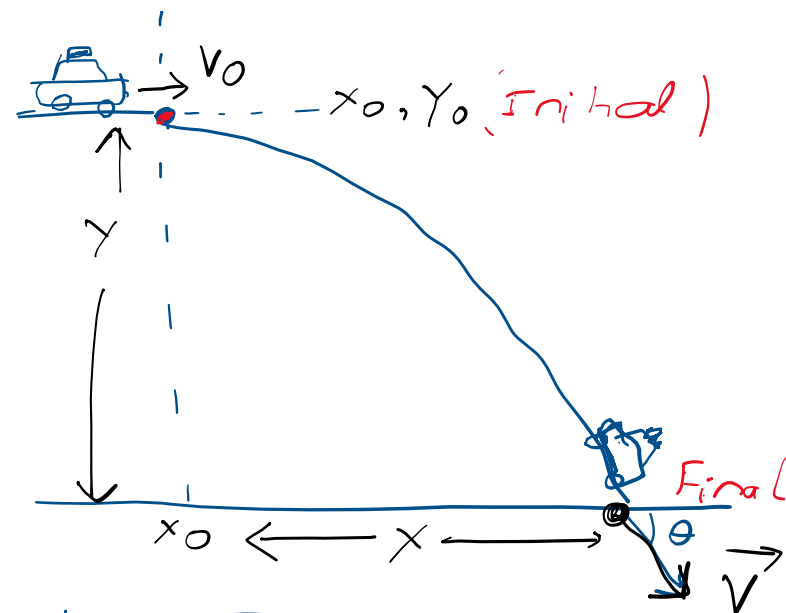
$$V_x = ?$$

$$V_y = ?$$

$$t = ?$$

$$\text{Finally, } |\vec{V}| = \sqrt{V_x^2 + V_y^2} = ?$$

$$\theta = \tan^{-1}\left(\frac{V_y}{V_x}\right) = ?$$



Equations for freefall, projectile motion

$$x = x_0 + V_{0x} t \quad \text{--- (1)}$$

$$V_y = V_{0y} - gt \quad \text{--- (2)}$$

$$Y = Y_0 + V_{0y} t - \frac{1}{2} g t^2 \quad \text{--- (3)}$$

$$V_y^2 = V_{0y}^2 - 2g(Y - Y_0) \quad \text{--- (4)}$$

Free fall, projectile motion

Example #4: Waymo

Analysis (We have information about $y = -20.0\text{m}$ and $x = 30.0\text{m}$)

Use equation (3)

(a)

$$y = \cancel{y_0} + \cancel{v_{0y}} t - \frac{1}{2} g t^2$$

$$y = -\frac{1}{2} g t^2$$

$$\sqrt{\frac{-2y}{g}} = t$$

$$t = \sqrt{\frac{-2(-20.0\text{m})}{(9.80)}}$$

$$\boxed{t = 2.02\text{s}}$$

(b) Now, the information of time from y -motion can be used for x -motion.

Use equation (2)

$$x = \cancel{x_0} + v_{0x} t$$

$$x = v_{0x} t$$

$$\frac{x}{t} = v_{0x}$$

$$v_{0x} = \frac{30.0\text{m}}{2.02\text{s}} = 14.8\text{m/s}$$

$$\boxed{v_0 = 14.8\text{m/s}}$$

Example #4: Waymo

© To find the final velocity ($\vec{v}$), we need both x-component (v_x) and y-component (v_y) of the final velocity.

$$v_x = v_o = 14.8 \text{ m/s}$$

$$v_y = v_{oy}^o - gt$$

$$v_y = -gt = -(9.80)(2.02) = -19.8 \text{ m/s}$$

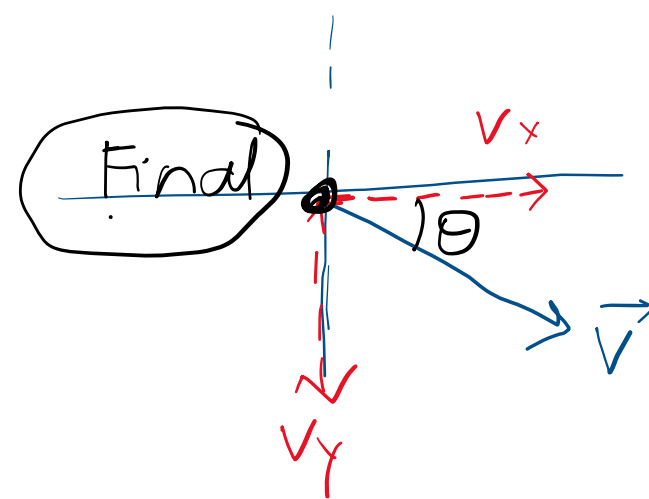
Now,

magnitude $|\vec{v}| = \sqrt{v_x^2 + v_y^2} = \sqrt{(14.8)^2 + (-19.8)^2} = \sqrt{611.08}$

$$|\vec{v}| = 24.7 \text{ m/s}$$

direction: $\theta = \tan^{-1}\left(\frac{v_y}{v_x}\right) = \tan^{-1}\left(\frac{-19.8}{14.8}\right)$

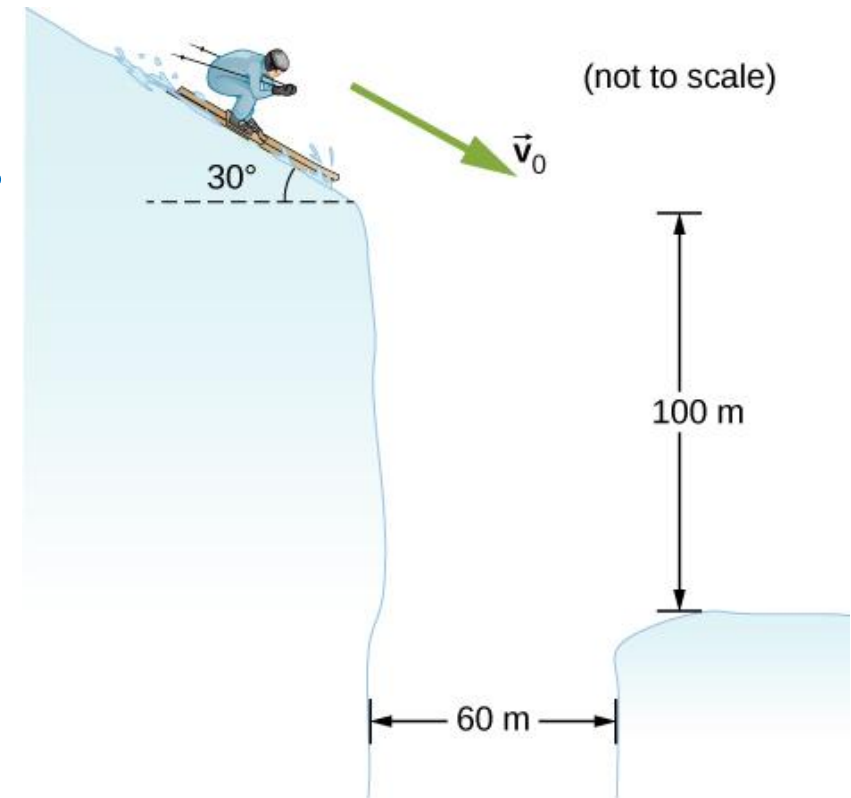
$$\theta = -53.2^\circ$$



Problem solving (Textbook #46)

Trying to escape his pursuers, a secret agent skis off a slope inclined at 30° below the horizontal at 16.68 m/s . To survive and land on the snow 100 m below, he must clear a gorge 60 m wide. Does he make it? Ignore air resistance.

Note: The y -component of $\vec{v}_0$ is in negative y -axis.



Given: $V_0 = 16.68 \text{ m/s}$

$$\theta_0 = 30^\circ$$

$$x_0 = 0 \text{ m}$$

$$y_0 = 0 \text{ m}$$

$$V_{0x} = V_0 \cos \theta_0 = 14.4 \text{ m/s}$$

$$V_{0y} = -V_0 \sin \theta_0 = -8.34 \text{ m/s}$$

Let's use the equation for position along y-direction

$$y = y_0 + V_{0y}t - \frac{1}{2}t^2$$

$$-100 = 0 - (8.34)t - \frac{1}{2}(9.80)t^2$$

$$-100 = -8.34t - 4.90t^2$$

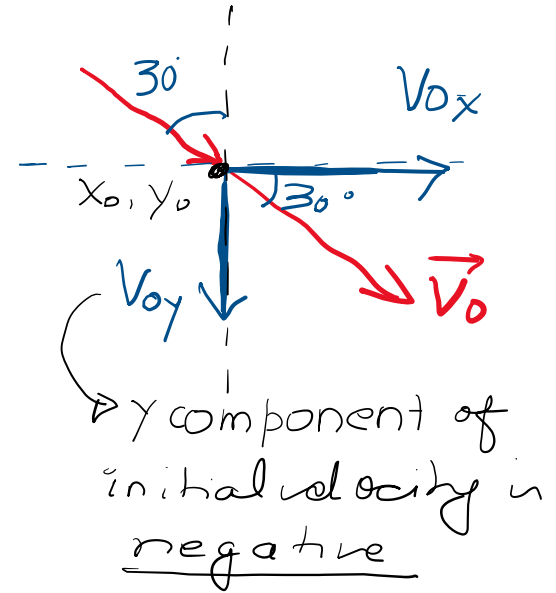
$$4.90t^2 + 8.34t - 100 = 0$$

$$ax^2 + bx + c = 0$$

Solve this quadratic equation in 't'

$$t = \frac{-8.34 \pm \sqrt{(8.34)^2 - 4(4.90)(100)}}{2(4.90)}$$

$$t = 3.75 \text{ s}$$



Problem solving (Textbook #46)

Now, use the time to find the distance along x-direction

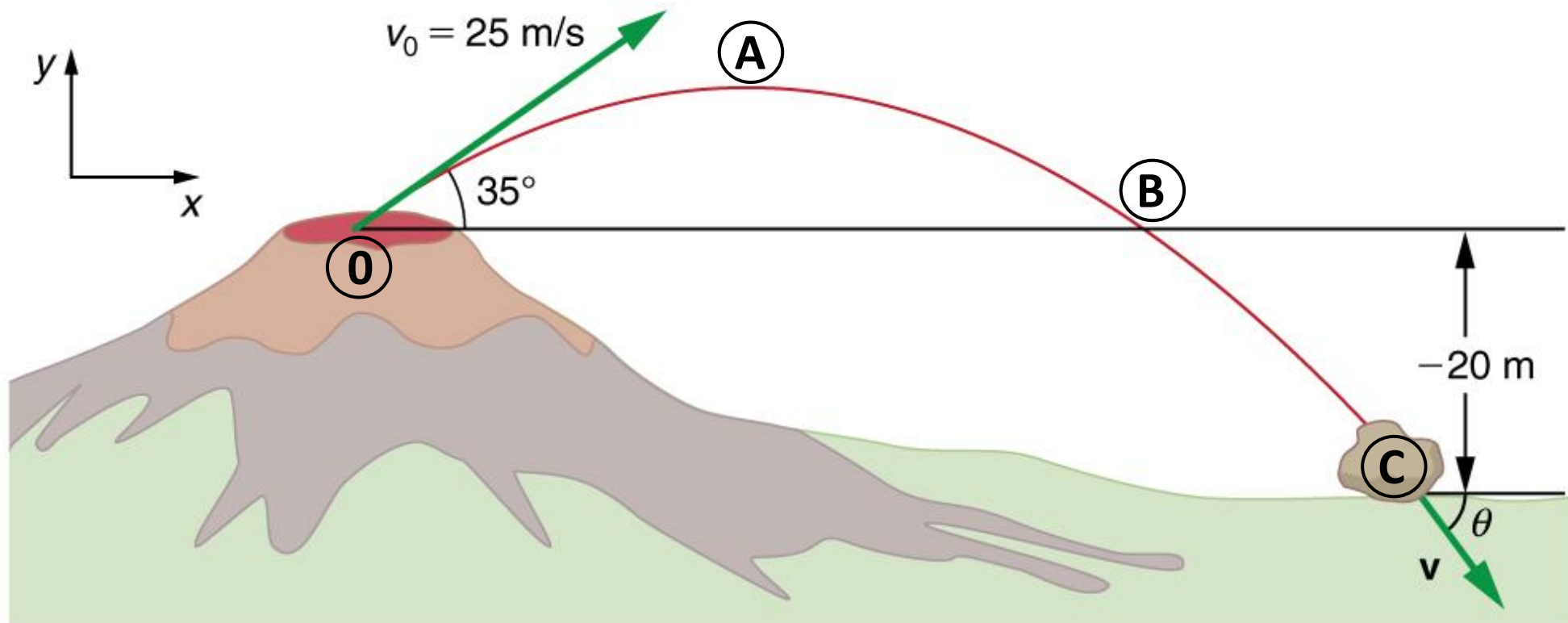
$$\begin{aligned}x &= x_0 + v_{0x} t \\ &= 0 + (14.4)(3.75)\end{aligned}$$

$$\boxed{x = 53.9 \text{ m}}$$

NO!! He did not make it.

Problem solving (Do it yourself !)

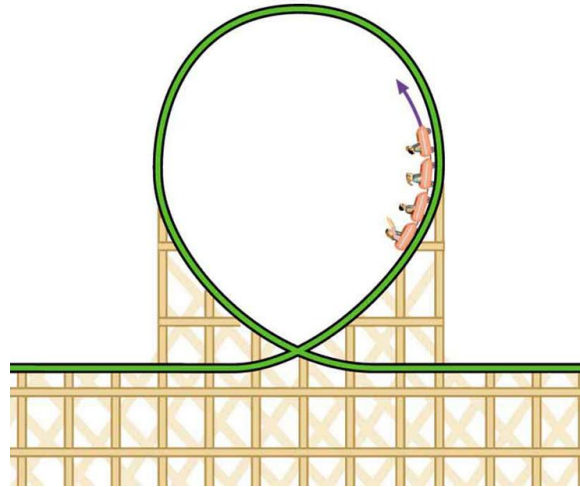
- a) Time when stone reaches to the ground (t_{OC})? [**3.96 s**]
- b) Magnitude and direction of velocity when the stones just hits the ground? [**31.9 m/s and -50.0°**]



The **trajectory** of a rock ejected from the Kilauea volcano.

Circular Motion

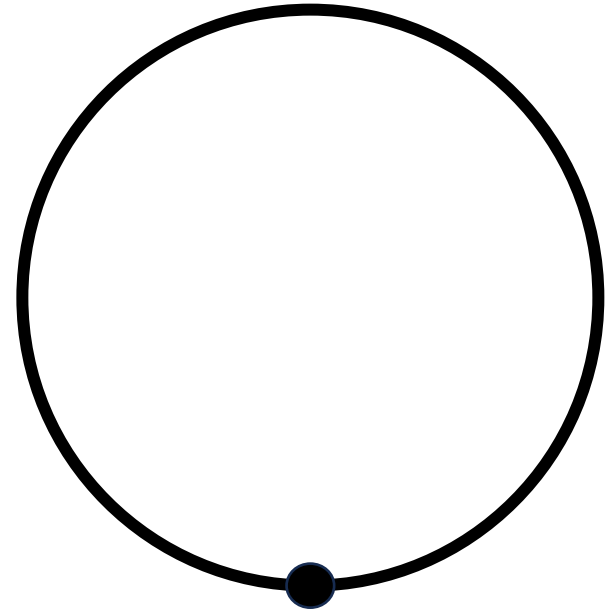
- Sounds like what it is, an object travelling in a circle
 - This includes objects spinning



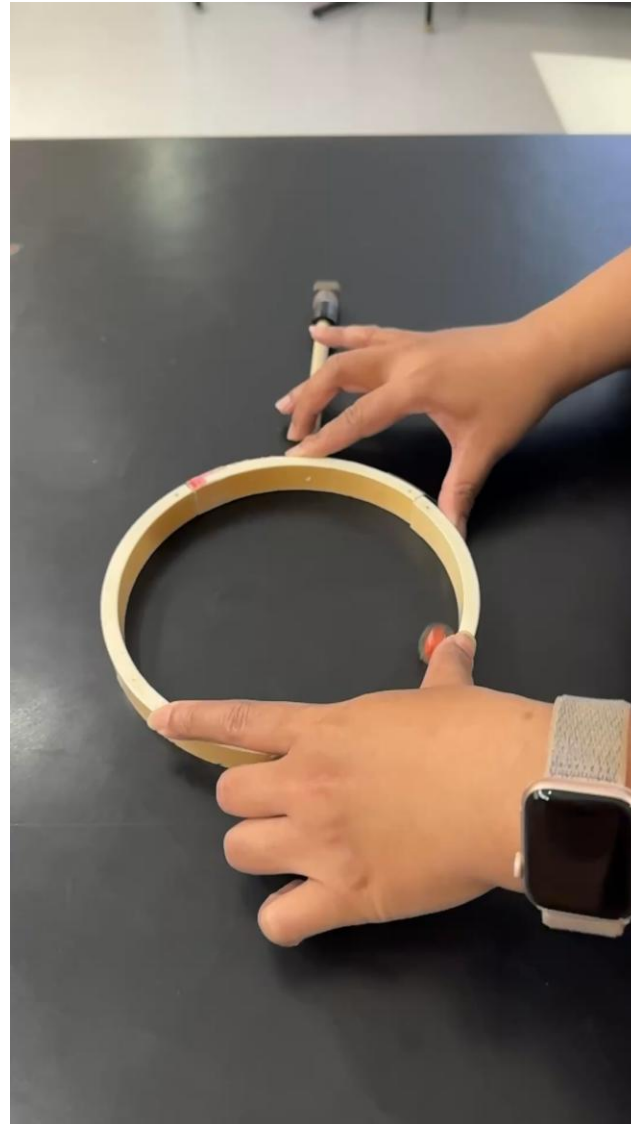
ABCD card

A ball is being swung **clockwise**. What is the direction of the tangential velocity at the point shown below?

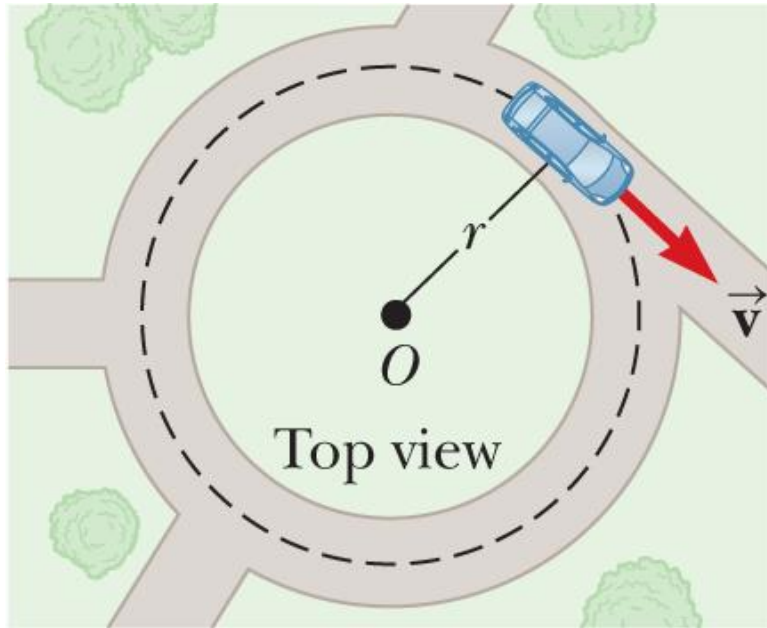
- a. Down
- b. Left
- c. Up
- d. Right



Uniform Circular Motion (Demo)



Uniform Circular Motion



a

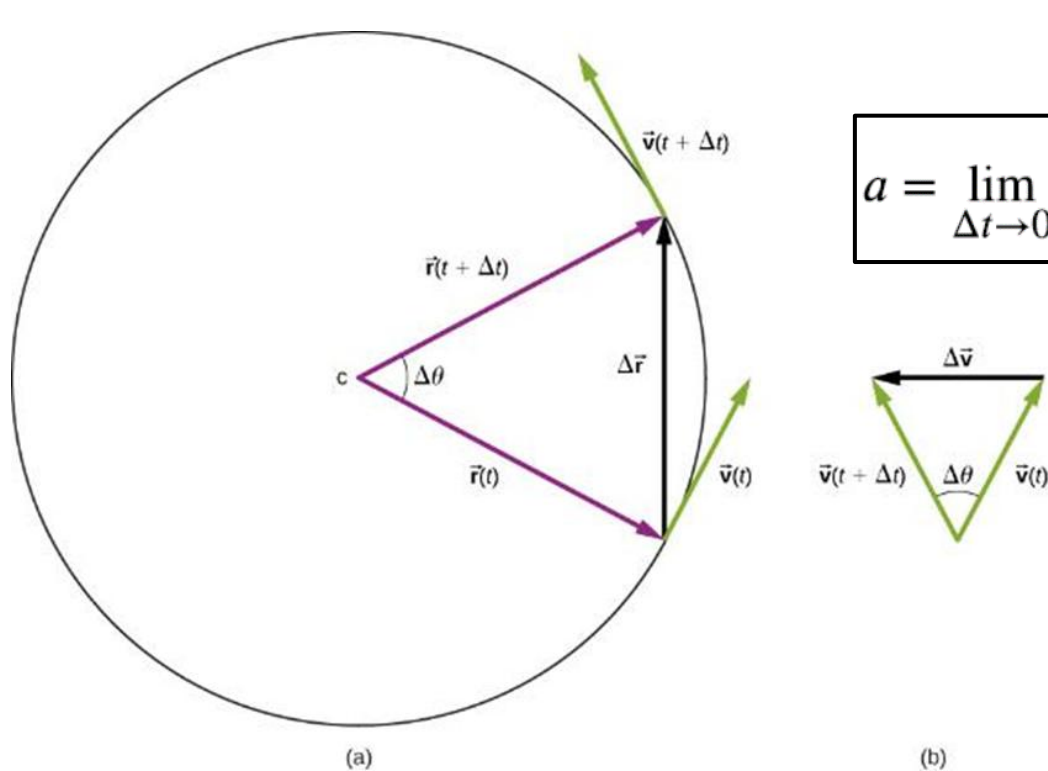
- If car moving with *constant speed* $v \rightarrow$ **uniform circular motion**.
- It still has an acceleration (change in the direction of velocity):

$$\vec{a} = \frac{d\vec{v}}{dt}$$

Conditions for Uniform Circular Motion

1. The tangential **speed**, v , is **constant**.
2. A body moving on a circular trajectory with *constant speed* is *always accelerated toward the center of the circle*.
3. Tangential **acceleration** is always **ZERO** in uniform circular motion.

Uniform Circular Motion



$$a = \lim_{\Delta t \rightarrow 0} \left(\frac{\Delta v}{\Delta t} \right) = \frac{v}{r} \left(\lim_{\Delta t \rightarrow 0} \frac{\Delta r}{\Delta t} \right) = \frac{v^2}{r}$$

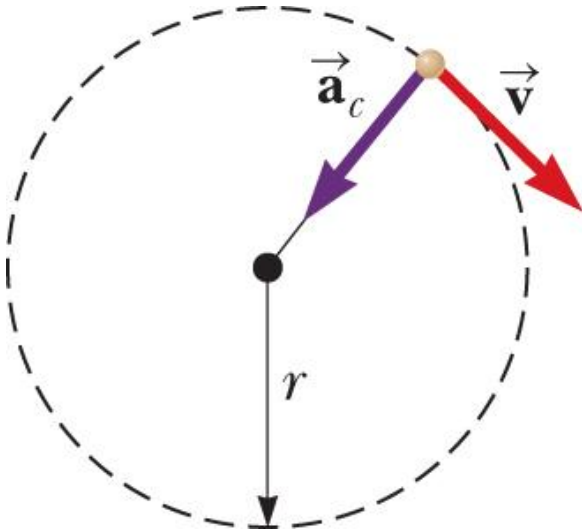
Similar triangles,

$$\frac{\Delta \vec{v}}{v} = \frac{\Delta \vec{r}}{r}$$

$$\Delta \vec{v} = \frac{v}{r} \Delta \vec{r}$$

- (a) A particle is moving in a circle at a constant speed, with position and velocity vectors at times t and $t + \Delta t$.
- (b) Velocity vectors forming a triangle. The two triangles in the figure are similar. The vector $\Delta \vec{v}$ points toward the center of the circle in the limit $\Delta t \rightarrow 0$.

Uniform Circular Motion



- Centripetal acceleration:

$$a_c = \frac{v^2}{r}$$

- Pointing toward center of the circle.

Radians

- Angle can be measured in degrees or radians!
- 1 revolution = 360 degrees = 2π radians
- Ratio of degrees to 360 equals the ratio of radians to 2π
 - $\frac{\theta_{deg}}{360} = \frac{\theta_{rad}}{2\pi}$
- Therefore: $\theta_{rad} = 2\pi \frac{\theta_{deg}}{360}$

CQ#3

How many **radians** is 45° ?

a. $\pi/4$

b. π

c. 2π

d. $\pi/8$

Arc length

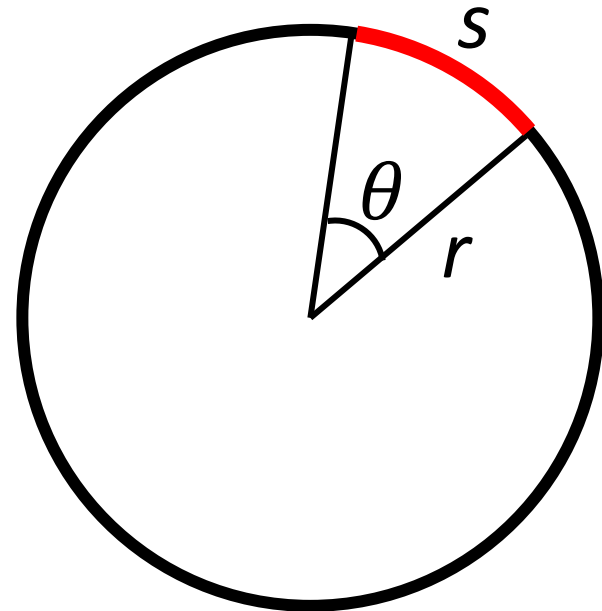
Definition: distance between two points along a section of a curve

Variable: s

Unit: m

Equation: $s = r\theta$

- r : Radius (m)
- θ : Angle (rad)



CQ#4

If you are walking on a circular path with radius r , what is the **distance** you travel after making half a revolution?

- a. $2r$
- b. $2\pi r$
- c. πr
- d. $360r$
- e. $180r$

Tangential velocity ($\vec{v}$)

Definition: Rate of change of translation

Variable: $\vec{v}$

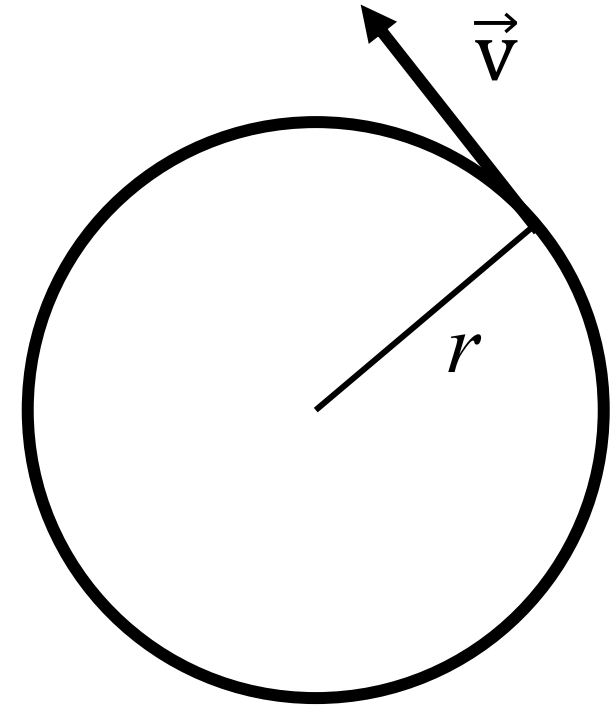
Unit: m/s

Equation: $\vec{v} = \frac{ds}{dt} = \frac{2\pi r}{T}$

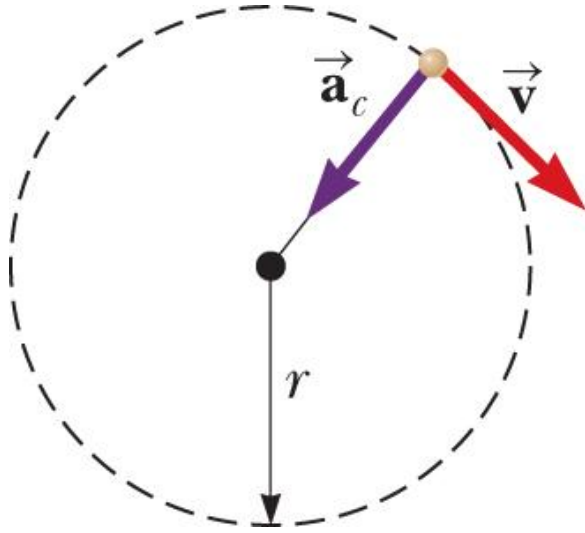
- r: radius (m)
- T: time for one revolution (s)

Direction: tangent to point of interest

- This is the direction in which object will go if released from the circular path



Uniform Circular Motion



- Centripetal acceleration:

$$a_c = \frac{v^2}{r}$$

- Pointing toward center of the circle.

$$v = \frac{2\pi r}{T} \rightarrow T = \frac{2\pi r}{v} \text{ (Period)}$$

Angular velocity (ω)

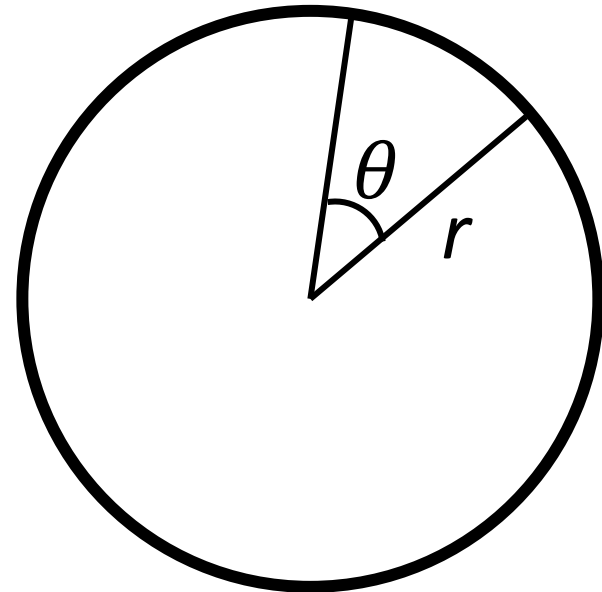
Definition: rate of change of angle

Variable: ω

Unit: rad/s

Equation: $\omega = \frac{\Delta\theta}{\Delta t}$

- In a full revolution: $\omega = \frac{2\pi}{T}$
 - T: Period,
i.e. time for one revolution



Connection between Tangential velocity and Angular velocity

$$v = \frac{dx}{dt} = \frac{ds}{dt} = \frac{d(r\theta)}{dt} = r \frac{d\theta}{dt} = r\omega$$

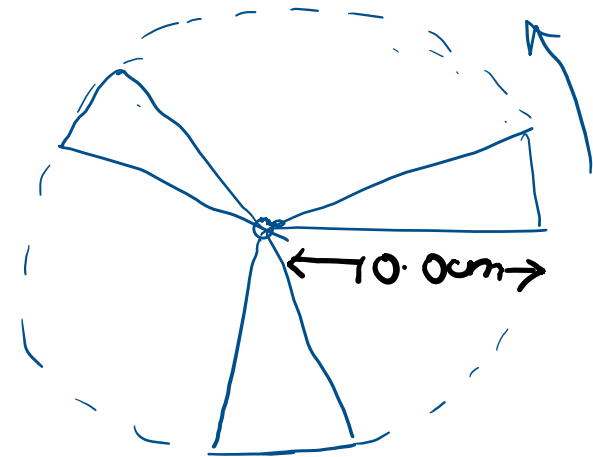
$$v = r\omega$$

$$a_c = \frac{v^2}{r} = r\omega^2$$

Problem Solving (Textbook#67)

1. A fan is rotating at a constant 360.0 rev/min. What is the magnitude of the acceleration of a point on one of its blades 10.0 cm from the axis of rotation?

Angular velocity (ω) = 360.0 rev/min



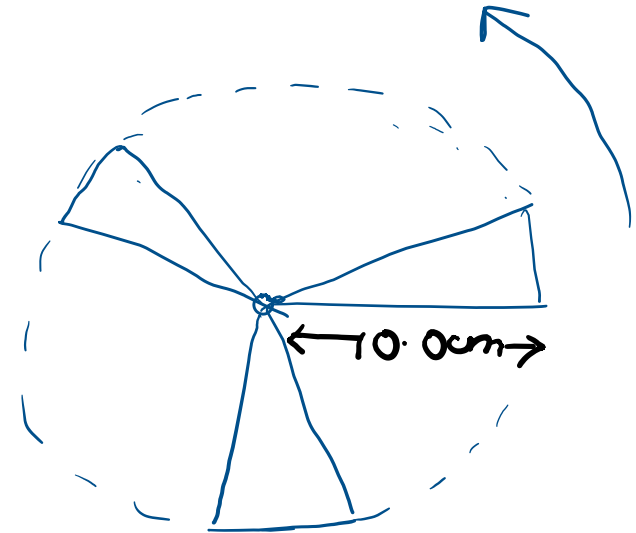
Problem Solving (Textbook#67)

1. A fan is rotating at a constant 360.0 rev/min. What is the magnitude of the acceleration of a point on one of its blades 10.0 cm from the axis of rotation?

Given;

$$\begin{aligned}\text{Angular velocity } (\omega) &= 360.0 \text{ rev/min} \\ &= 360.0 \frac{\cancel{\text{rev}}}{\cancel{\text{min}}} \times \frac{2\pi \text{ rad}}{1 \cancel{\text{rev}}} \times \frac{1 \cancel{\text{min}}}{60 \text{ s}}\end{aligned}$$

$$\omega = 37.7 \text{ rad/s}$$



Problem Solving (Textbook#67)

$$\omega = 37.7 \text{ rad/s}$$

$$r = 10.0 \text{ cm} = 10.0 \cancel{\text{ cm}} \times \frac{1 \text{ m}}{100 \cancel{\text{ cm}}} = 0.100 \text{ m}$$

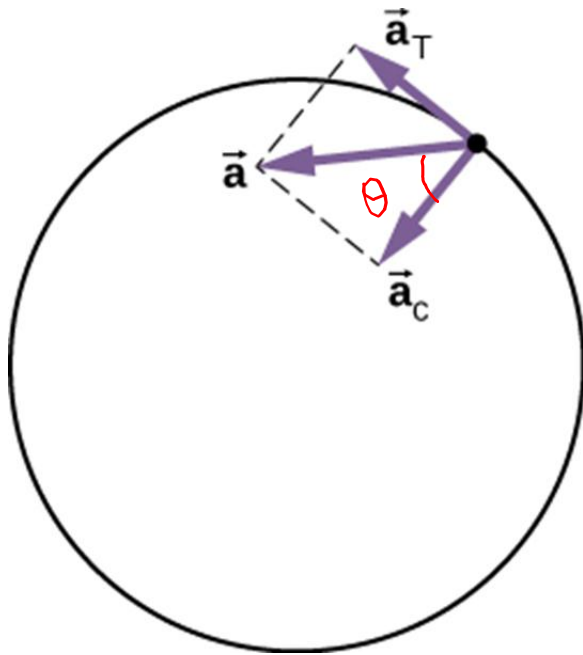
Now, the centripetal acceleration,

$$a_c = \omega^2 r$$

$$= (37.7)^2 (0.100)$$

$$\boxed{a_c = 142 \text{ m/s}^2}$$

Non-Uniform Circular Motion



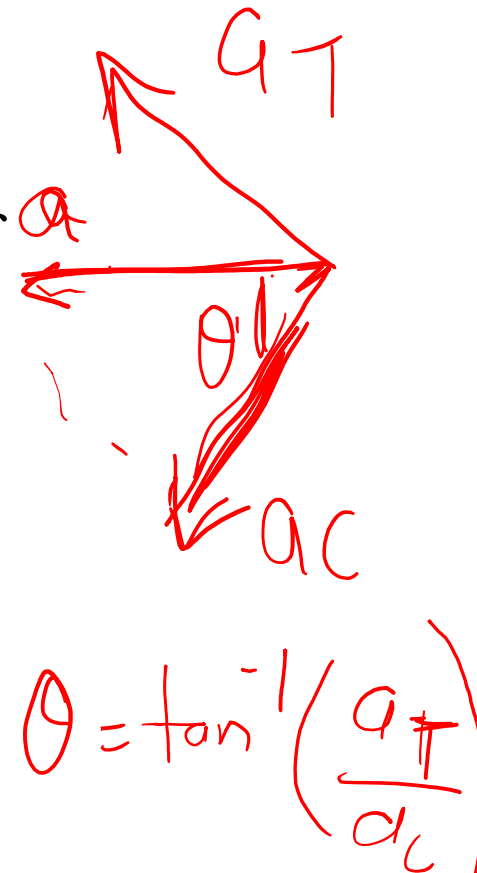
Centripetal acceleration: the time rate of change of the *direction* of the velocity.

$$a_c = \frac{v^2}{r}.$$

Tangential acceleration: the time rate of change of the *magnitude* of the velocity.

$$a_T = \frac{d|\vec{v}|}{dt}.$$

Total acceleration: $\vec{a} = \vec{a}_c + \vec{a}_T.$ $a = \sqrt{a_c^2 + a_T^2}$



Chapter 4 Summary: Concepts

Position vector

$$\vec{\mathbf{r}}(t) = x(t)\hat{\mathbf{i}} + y(t)\hat{\mathbf{j}} + z(t)\hat{\mathbf{k}}$$

Displacement vector

$$\Delta\vec{\mathbf{r}} = \vec{\mathbf{r}}(t_2) - \vec{\mathbf{r}}(t_1)$$

Velocity vector

$$\vec{\mathbf{v}}(t) = \lim_{\Delta t \rightarrow 0} \frac{\vec{\mathbf{r}}(t+\Delta t) - \vec{\mathbf{r}}(t)}{\Delta t} = \frac{d\vec{\mathbf{r}}}{dt}$$

Velocity in terms of components

$$\vec{\mathbf{v}}(t) = v_x(t)\hat{\mathbf{i}} + v_y(t)\hat{\mathbf{j}} + v_z(t)\hat{\mathbf{k}}$$

Velocity components

$$v_x(t) = \frac{dx(t)}{dt} \quad v_y(t) = \frac{dy(t)}{dt} \quad v_z(t) = \frac{dz(t)}{dt}$$

Average velocity

$$\vec{\mathbf{v}}_{\text{avg}} = \frac{\vec{\mathbf{r}}(t_2) - \vec{\mathbf{r}}(t_1)}{t_2 - t_1}$$

Instantaneous acceleration

$$\vec{\mathbf{a}}(t) = \lim_{t \rightarrow 0} \frac{\vec{\mathbf{v}}(t+\Delta t) - \vec{\mathbf{v}}(t)}{\Delta t} = \frac{d\vec{\mathbf{v}}(t)}{dt}$$

Instantaneous acceleration, component form

$$\vec{\mathbf{a}}(t) = \frac{dv_x(t)}{dt}\hat{\mathbf{i}} + \frac{dv_y(t)}{dt}\hat{\mathbf{j}} + \frac{dv_z(t)}{dt}\hat{\mathbf{k}}$$

Instantaneous acceleration as second derivatives of position

$$\vec{\mathbf{a}}(t) = \frac{d^2x(t)}{dt^2}\hat{\mathbf{i}} + \frac{d^2y(t)}{dt^2}\hat{\mathbf{j}} + \frac{d^2z(t)}{dt^2}\hat{\mathbf{k}}$$

Chapter 4 Summary: 2D Motion Kinematics

- Motions along **perpendicular** axes are independent and thus can be analyzed separately.
- Consider a 2D problem of a particle moving in the xy plane with constant acceleration. The acceleration vector is:

$$\vec{\mathbf{a}} = a_{0x}\hat{\mathbf{i}} + a_{0y}\hat{\mathbf{j}}.$$

- Equations for position and velocity in the x and y -directions:

$$x(t) = x_0 + (v_x)_{\text{avg}}t$$

$$y(t) = y_0 + (v_y)_{\text{avg}}t$$

$$v_x(t) = v_{0x} + a_x t$$

$$v_y(t) = v_{0y} + a_y t$$

$$x(t) = x_0 + v_{0x}t + \frac{1}{2}a_x t^2$$

$$y(t) = y_0 + v_{0y}t + \frac{1}{2}a_y t^2$$

$$v_x^2(t) = v_{0x}^2 + 2a_x(x - x_0)$$

$$v_y^2(t) = v_{0y}^2 + 2a_y(y - y_0).$$

Chapter 4 Summary: Projectile Motion

- **Horizontal** and **vertical** components of projectile's motion are completely independent and can be handled separately:
- Horizontal direction: constant velocity motion ($a_x = 0$);
- Vertical direction: constant acceleration motion ($a_y = -g$).

Horizontal Motion

$$V_{0x} = V_0 \cos \theta_0$$

$$v_{0x} = v_x, \quad x = x_0 + v_x t$$

Vertical Motion

$$V_{0y} = V_0 \sin \theta_0$$

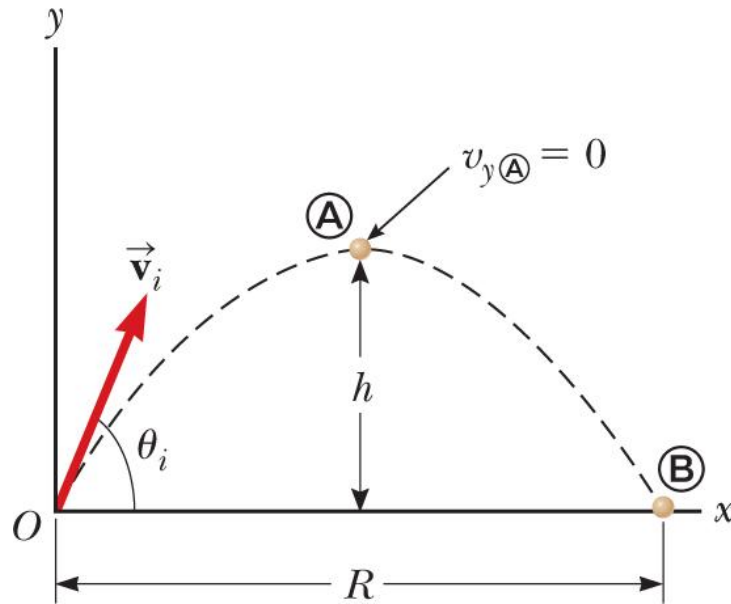
$$y = y_0 + \frac{1}{2}(v_{0y} + v_y)t$$

$$v_y = v_{0y} - gt$$

$$y = y_0 + v_{0y}t - \frac{1}{2}gt^2$$

$$v_y^2 = v_{0y}^2 - 2g(y - y_0)$$

Chapter 4 Summary: Projectile Motion



$$t_A = \frac{v_0 \sin \theta_0}{g}$$

$$T_{tof} = 2 t_A = \frac{2 v_0 \sin \theta_0}{g}$$

$$h = \frac{v_0^2 \sin^2 \theta_0}{2g}$$

$$R = \frac{v_0^2 \sin 2 \theta_0}{g}$$

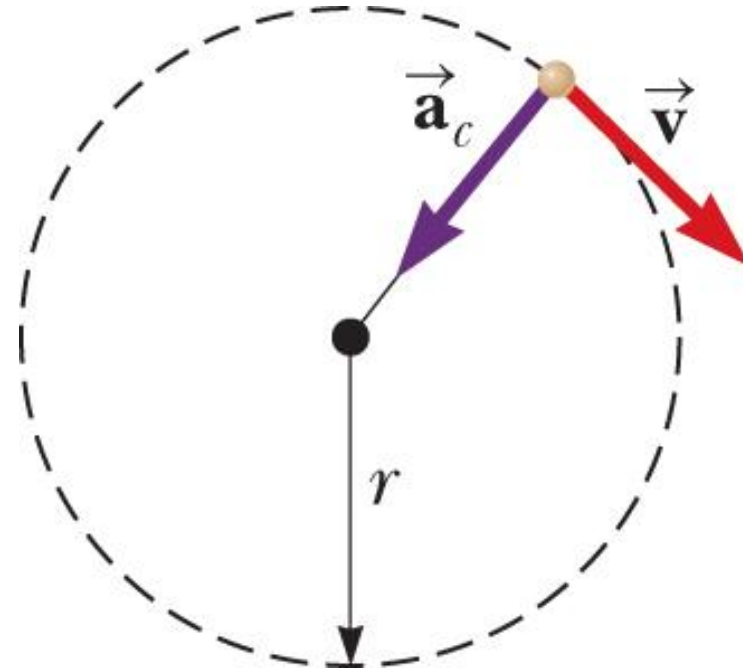
Chapter 4 Summary: Uniform Circular Motion

Centripetal acceleration:

$$a_c = \frac{v^2}{r}$$

Period: $T = \frac{2\pi r}{v}$

Angular frequency: $\omega = \frac{2\pi}{T}$



Check Your Learning at Three Levels

1. Physics Concepts and Definitions

2D displacement, velocity, speed, acceleration

2. Physics Laws and Formulas

2D motion kinematics equations (constant a)

3. Use Formulas to Solve Problems

2D motion with constant a , projectile motion